GB

Installation details

for rolling shutters, rolling gratings and perforated gratings

Date: 08.2007

Teckentrup

Türen · Tore · Zargen

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2. General advice

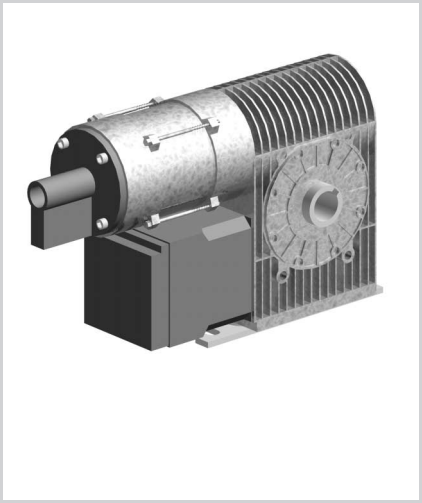
- The example for installation shows the right-hand version of the operator, left-hand version available on request.
- Please take the minimum headroom heights from the tables starting on page 30.
- For outside rolling shutters the operator side is determined from the outside.
- Measurement indications for special constructions respectively special wall connections - if the necessary space requirements are not sufficient - on request.
- Subject to technical changes.

3. Technical terms and index of abbreviations

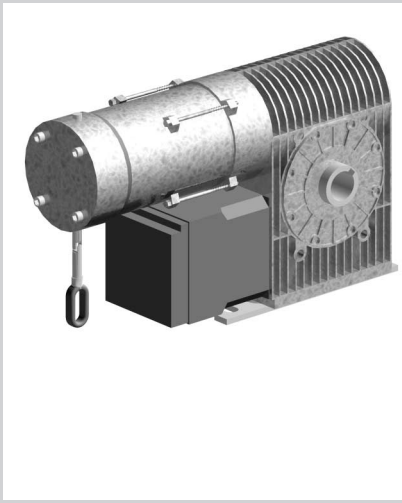
AF: Sideroom safety device
AK: Sideroom chain drive with emergency crank handle (standard)
AS: Sideroom direct drive with emergency crank handle (standard)
BB: Order dimensions width
BH: Order dimensions height
BS: Pass width
ET: Installation depth (see page 30)
FS: Sideroom guide rail
GRAM: Total RAM (overall frame size)
H: Height
KE: Chain (emergency chain drive)
KKU: Universal joint crank (folding)
KU: Crank (emergency crank handle)
L: Length
LDB: Clear passage width
LDH: Clear passage height
LS: Sideroom on bearing side
N: Eye for awning rod
NWT: Emergency operation
OFF: Upper edge finished floor
OL: Top light
RAM: Overall frame size
RRM: Theoretical dimension in unfinished state
S: Headroom (see page 30)
TAR: Teckentrup direct mount drive roller shutter
TKR: Teckentrup chain drive roller shutter
TRR: Teckentrup tubular motor roller shutter
WWV: Tube extension

4. Emergency operations

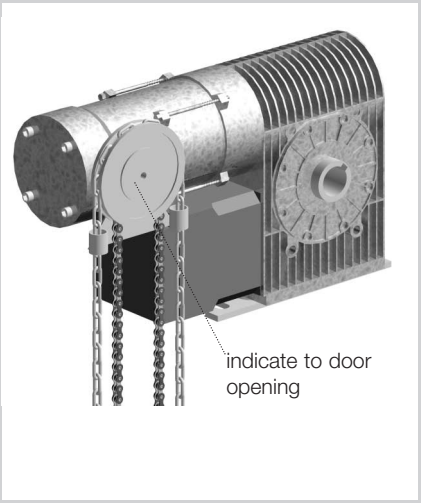
Direct drive operator with emergency crank handle (KU)



Direct drive operator with awning rod (NWT)



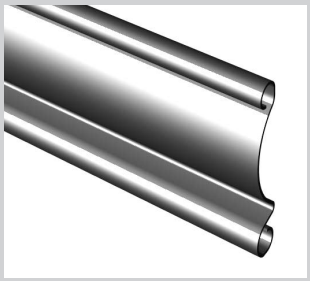
Direct drive operator with hand chain (KE)

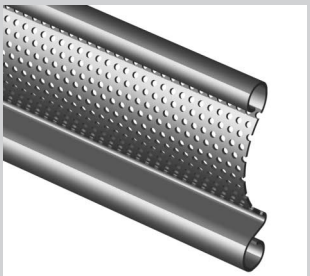


Images may differ from actual product.

5. Profile cross-sections (rolling shutter profiles)

Rolling shutter profiles
- single-skinned



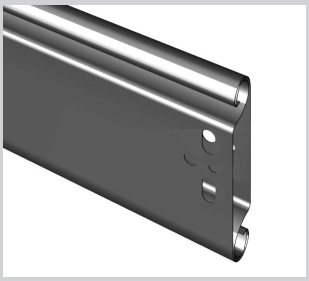


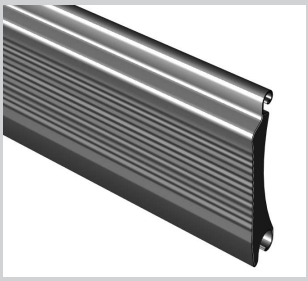
Profiles	Profile type	6010	6010 visio
	Insulating	No	No
	No. of rods per m door height	10,7	10,7
	Profile height in mm	104	104
	Profile partition in mm	93,2	93,2
Weights per square meter	at 0,4 mm	----	----
	at 0,8 mm	----	----
	at 1,0 mm	Al 4,3 kg / pcs 13,0 kg	Al 3,7 kg
	at 1,2 mm	Al 5,2 kg	----
Version	Inside roller / outside roller	Yes / Yes③	Yes / Yes③
	Window measurement	----	----
	Smallest shaft diameter	ø 133	ø 133
	Fitting door seal	6013	6013
	Window / Ventilation	No / No	No / Yes (31% per square meter)
	Insulating values	----	----
Steel galvanized	Material thickness	1,0	----
	Max. door width in mm	7500②	----
Aluminium	Material thickness	1,0 / 1,2	1,0
	Max. door width in mm	5000 / 6500③	5500

Advice:
In accordance with EN 13241-1 shutters installed in the facade must have at least resistance to wind load Class 2.
Exception: Indoor shutters or shutters installed in fully wind-protected openings.

5. Profile cross-sections (rolling shutter profiles)

Rolling shutter profiles
- double-skinned



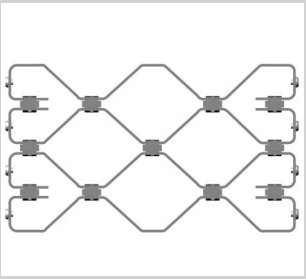


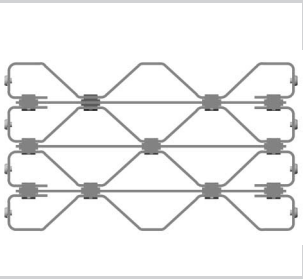
Profiles	Profile type	4020	ThermoTeck
	Insulating	Styrofoam filling	PU foamed
	No. of rods per m door height	10	10,7
	Profile height in mm	116,5	106
	Profile partition in mm	100	93,7
Weights per square meter	at 0,4 mm	----	St 11,3 kg
	at 0,8 mm	Al 7,6 kg	----
	at 1,0 mm	Al 9,5 kg	----
	at 1,2 mm	----	----
Version	Inside roller / outside roller	Yes / Yes	Yes / Yes④
	Window measurement	150x50 / 5 pcs/m	150x55 / 5 pcs/m
	Smallest shaft diameter	ø 159	ø 133
	Fitting door seal	4013	ET-1
	Window / Ventilation	Yes / No	Yes / Yes (11% per square meter)
	Insulating values	k-value ~4,3 W/m²K (Al) RW-value: ~21 dB	k-value ~3,5 W/m²K (St) k-value ~4,0 W/m²K (Al) RW-value: ~20 dB
Steel galvanized	Material thickness	----	0,4
	Max. door width in mm	----	9000②
Aluminium	Material thickness	0,8 / 1,0	----
	Max. door width in mm	3750 / 9000①	----

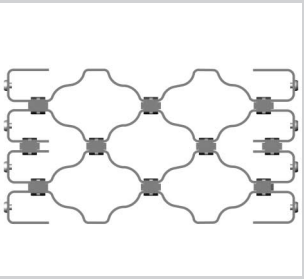
① in case of lower wind pressure, door width up to 15.000 mm is possible.
② in case of lower wind pressure, door width up to 12.000 mm is possible.
③ With outdoor (reverse coil) roller shutters, the actual inside is visible from the outside. The curve faces the inside.
④ With outdoor (reverse coil) roller shutters, the actual inside is visible from the outside. The micro-profiling faces the inside.

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6. Honeycomb cross-sections (rolling shutter honeycombs)

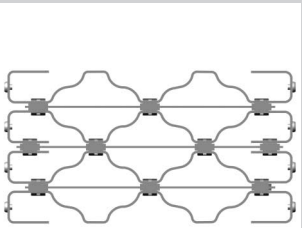


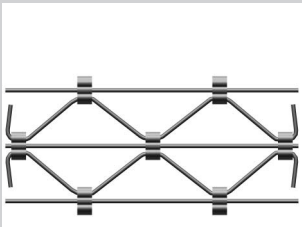




Grating	Profile type	Straight honeycomb 185		Straight honeycomb 185 with cross rod		Wave honeycomb 160	
	Description	TG-		TG-		TG-	
	Version	**S**teel / Fine steel (**E**) / **A**luminium		**S**teel / Fine steel (**E**) / **A**luminium		**S**teel / Fine steel (**E**) / **A**luminium	
	No. of honeycombs per m door height	~7,2		~7,2		~8,0	
	Honeycomb height in mm	~125		~125		~110	
	Honeycomb partition in mm	~138		~138		~124	
Weights per square meter	at 16x4	Al 5,5 kg / St 14,0 kg		Al 8,0 kg / St 22,0 kg		Al 5,5 kg / St 14,0 kg	
	at 20x4	Al 6,5 kg / St 16,0 kg		Al 9,0 kg / St 24,0 kg		Al 6,5 kg / St 16,0 kg	
	Oval rod 11x7	----		----		----	
	Smallest shaft diameter	ø 133		ø 133		ø 133	
	Bottom seal	Steel angle		Steel angle		Steel angle	
	Ventilation area	approx. 85%		approx. 82%		approx. 85%	
Steel / Fine steel	Material thickness	16x4	20x4	16x4	20x4	16x4	20x4
	Max. door width in mm	4.500	11.000	4.500	11.000	4.500	11.000
Aluminium	Material thickness	16x4	20x4	16x4	20x4	16x4	20x4
	Max. door width in mm	4.000	11.000	4.000	11.000	4.000	11.000

6. Honeycomb cross-sections (rolling shutter honeycombs)

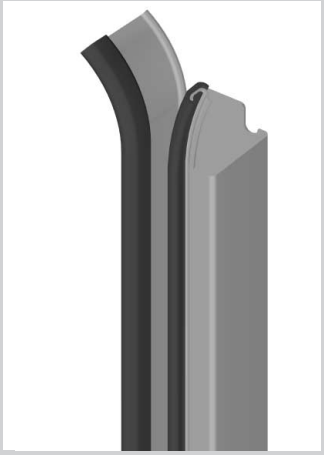




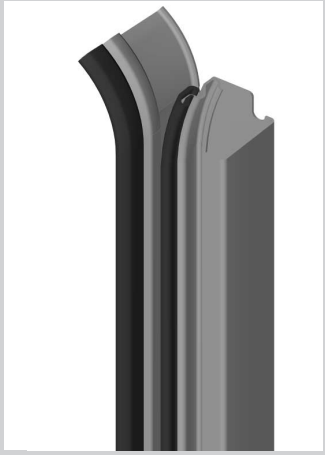
Grating	Profile type	Wave honeycomb 160 with cross rod	Straight honeycomb 225 with cross rod
	Description	TG-	TG-X
	Version	**S**teel / Fine steel (**E**) / **A**luminium	**A**luminium
	No. of honeycombs per m door height	~8,0	~5,2
	Honeycomb height in mm	~110	~170
	Honeycomb partition in mm	~124	~190
Weights per square meter	at 16x4	Al 8,0 kg / St 22,0 kg	----
	at 20x4	Al 9,0 kg / St 24,0 kg	----
	Oval rod 11x7	----	Al 5,8 kg
	Smallest shaft diameter	ø 133	ø 133
	Bottom seal	Steel angle	Steel angle
	Ventilation area	approx. 82%	approx. 76%
Steel / Fine steel	Material thickness	16x4 / 20x4	----
	Max. door width in mm	4.500 / 11.000	----
Aluminium	Material thickness	16x4 / 20x4	11x7
	Max. door width in mm	4.000 / 11.000	11.000

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7. Determination of guide rails



FS 85 steel galvanized



FS 110 steel galvanized

Advice:
For safety catches in general FS 110.

Profile	Version	Wind pressure	Door size in mm FS 85	Door size in mm FS 110	Use of safety catches
6010	Aluminium	450 Pa	up to 4,250	from 4,250	from 4,750
		700 Pa	up to 3,750	from 3,750	from 4,250
		1,000 Pa	up to 3,250	from 3,250	from 3,750
	Steel	450 Pa	up to 5,000	from 5,000	from 5,500
		700 Pa	up to 4,000	from 4,000	from 4,500
		1,000 Pa	up to 3,750	from 3,750	from 4,000
4020	Aluminium	450 Pa	up to 5,750	from 5,750	from 6,250
		700 Pa	up to 4,750	from 4,750	from 5,250
		1,000 Pa	up to 4,000	from 4,000	from 4,750
ThermoTeck	Steel	450 Pa	up to 5,500	from 5,500	from 6,000
		700 Pa	up to 4,500	from 4,500	from 5,000
		1,000 Pa	up to 4,000	from 4,000	from 4,250
Rolling gratings	Aluminium	--	up to 5,500	from 5,500	--
	Steel	--	up to 6,000	from 6,000	--

Wind pressure conversion

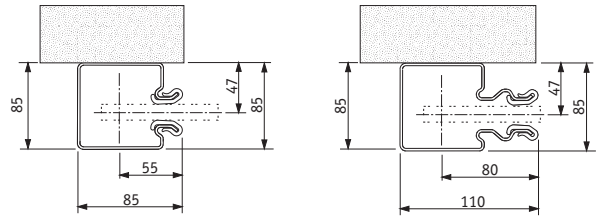
Wind pressure	**Resistance to wind load class**	**Pa (Pascal)**	**daN/m²**	**km/h**	**m/s**
	1	300	30	79	22
	2	450	45	96	27
	3	700	70	120	33
	4	1.000	100	144	40

Our roller shutters correspond to resistance to wind load class 2 in accordance with DIN EN 13241-1.
Other resistance to wind load classes are available.

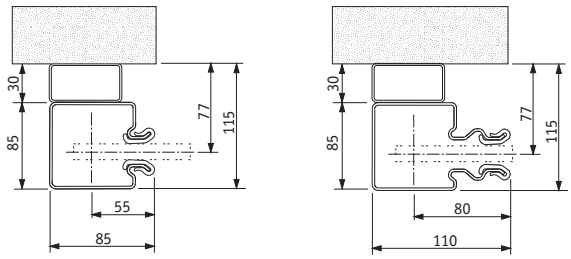
8. Guide rail profiles

FS 85 FS 110

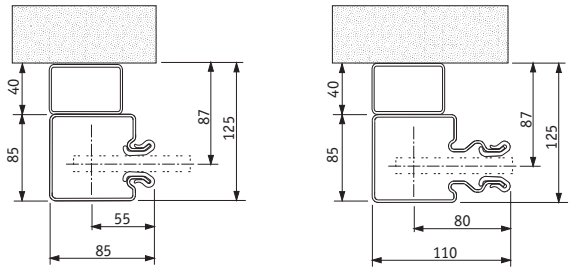
Without lining



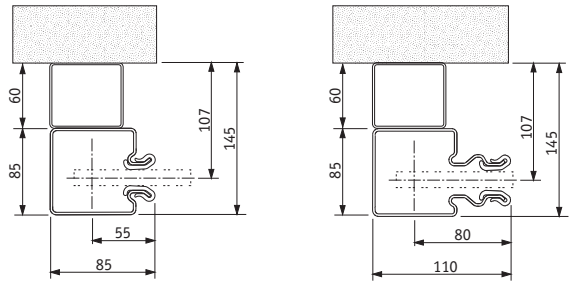
Lining 30 mm



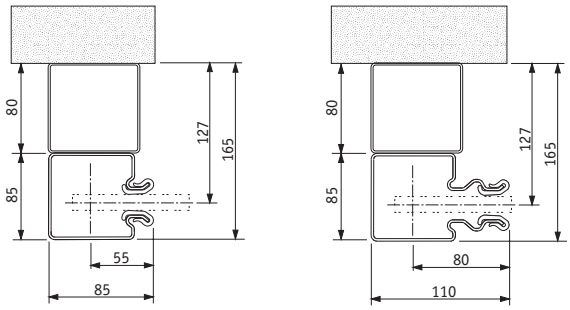
Lining 40 mm*



Lining 60 mm



Lining 80 mm





Advice:

- For height of lining tube please look up starting from page 30.
- The guide rail is equipped by factory with a profiled lining.
- FS 85 only for doors without safety catches (for door sizes please look up page 8).
- FS 110 for doors without / with safety catches (for door sizes please look up page 8).

* Underfill 40 mm as standard for roller grilles with impulse control MS 700 and/or shutter width > 5,000 mm.

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9. Clearance direct drive

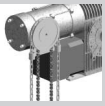
Required fitting widths

Guide rail type:		FS 85	FS 110
FS		85	110
LS		175	200
AS with KU	TAR 1.14	300	325
	TAR 2.22	300	325
	TAR 3.30	300	325
	TAR 3.42	300	325
	TAR 3.50	300	325
	TAR 5.75	300	325
	TAR 6.100	315	340
AS with KE	TAR 1.14	240	265
with KKU	TAR 2.22	240	265
with NWT	TAR 3.30	240	265
	TAR 3.42	240	265
	TAR 3.50	-	-
	TAR 5.75	-	-
	TAR 6.100	-	-

KU = crank (emergency crank handle)



KE = chain (emergency chain drive)



KKU = universal joint crank (folding)



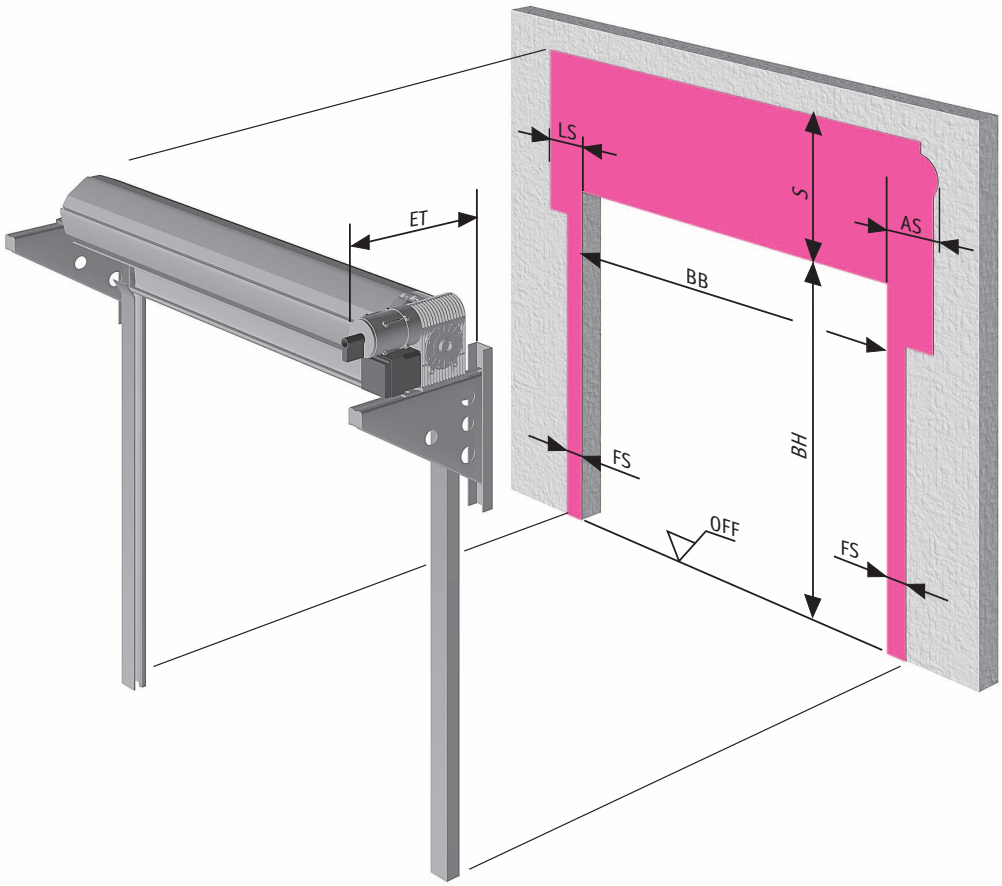
NWT = Emergency operation (with awning rod) only up to TAR 2.22



- FS (dimension requirement) see page 8
- Drive (dimension requirement) see page 30

All measurements in mm

Mounting on any wall material.



10. Clearance chain drive

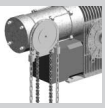
Required fitting widths

Guide rail type:		FS 85	FS 110
FS		85	110
LS		225	250
AS with KU	TKR 1.10	310	335
	TKR 2.15	310	335
	TKR 3.22	310	335
	TKR 4.30	325	350
	TKR 5.40	325	350
	TKR 6.60	435	460
	TKR 7.97	435	460
AS with KE	TKR 1.10	310	335
	TKR 2.15	310	335
	TKR 3.22	310	335
	TKR 4.30	-	-
	TKR 5.40	-	-
	TKR 6.60	-	-
	TKR 7.97	-	-

KU = crank (emergency crank handle)



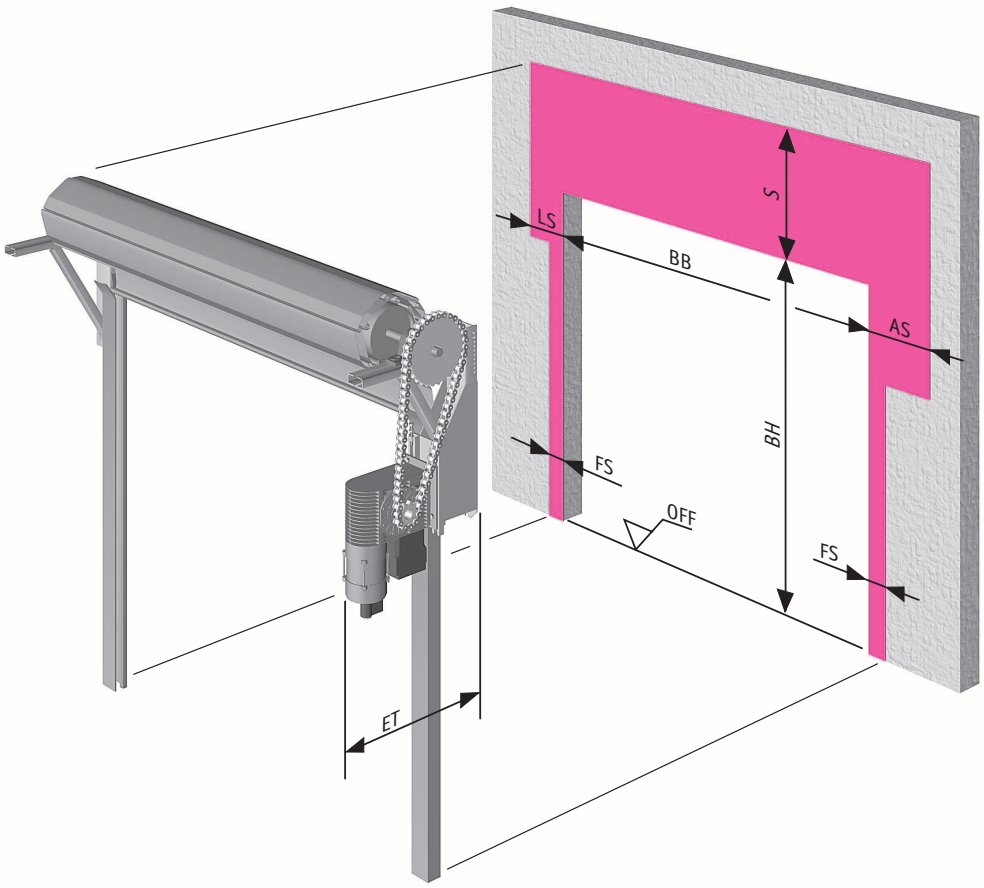
KE = chain (emergency chain drive)



- FS (dimension requirement) see page 8
- Drive (dimension requirement) see page 30

All measurements in mm

Mounting on any wall material.



11. Clearance direct drive and pressure shaft

Required fitting widths

Guide rail type:		FS 85	FS 110
FS		85	110
LS		275	300
AS with KU	TAR 1.14	400	425
	TAR 2.22	400	425
	TAR 3.30	400	425
	TAR 3.42	400	425
	TAR 3.50	400	425
	TAR 5.75	400	425
	TAR 6.100	415	440
AS with KE	TAR 1.14	340	365
with KKU	TAR 2.22	340	365
with NWT	TAR 3.30	340	365
	TAR 3.42	340	365
	TAR 3.50	-	-
	TAR 5.75	-	-
	TAR 6.100	-	-

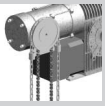
All measurements in mm

Mounting on any wall material.

KU = crank (emergency crank handle)



KE = chain (emergency chain drive)



KKU = universal joint crank (folding)

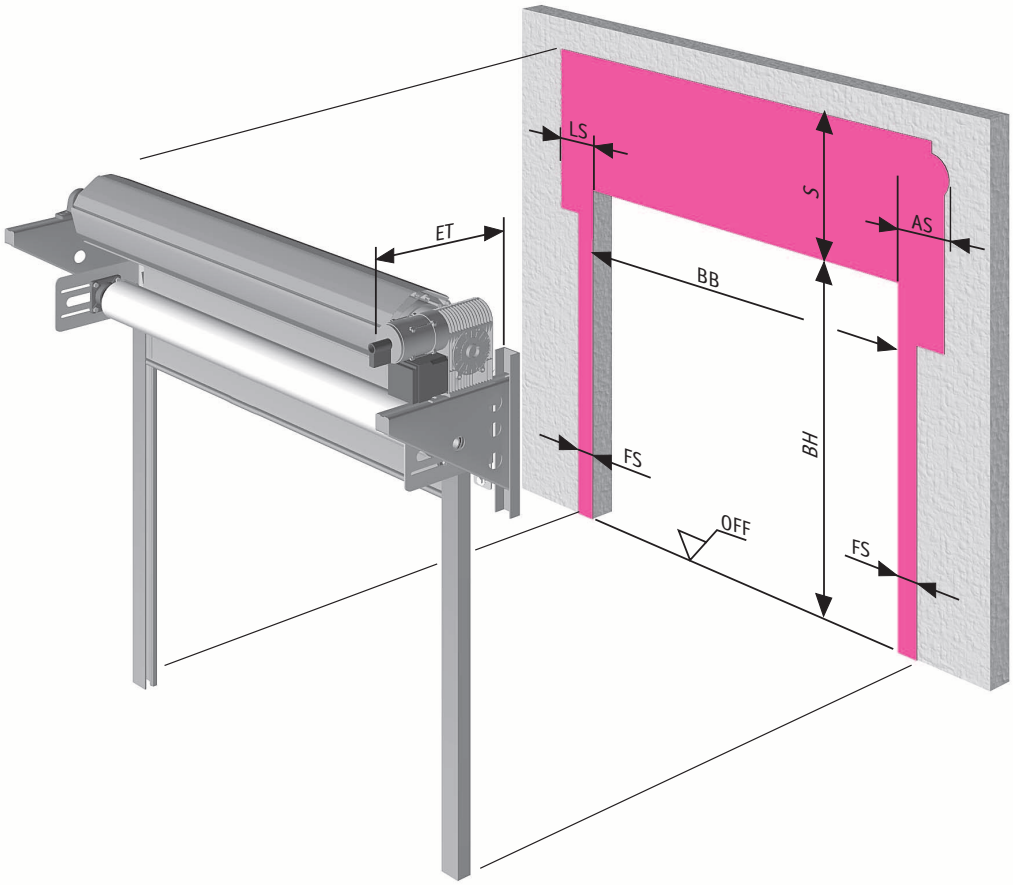


NWT = Emergency operation (with awning rod) only up to TAR 2.22



- FS (dimension requirement) see page 8
- Drive (dimension requirement) see page 30

The headroom height S is increased by 210 mm for doors with press shaft.



12. Clearance chain drive and pressure shaft

Required fitting widths

Guide rail type:		FS 85	FS 110
FS		85	110
LS		325	350
AS with KU	TKR 1.10	410	435
	TKR 2.15	410	435
	TKR 3.22	410	435
	TKR 4.30	425	450
	TKR 5.40	425	450
	TKR 6.60	535	560
	TKR 7.97	535	560
AS with KE	TKR 1.10	410	435
	TKR 2.15	410	435
	TKR 3.22	410	435
	TKR 4.30	-	-
	TKR 5.40	-	-
	TKR 6.60	-	-
	TKR 7.97	-	-

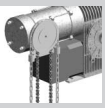
All measurements in mm

Mounting on any wall material.

KU = crank (emergency crank handle)

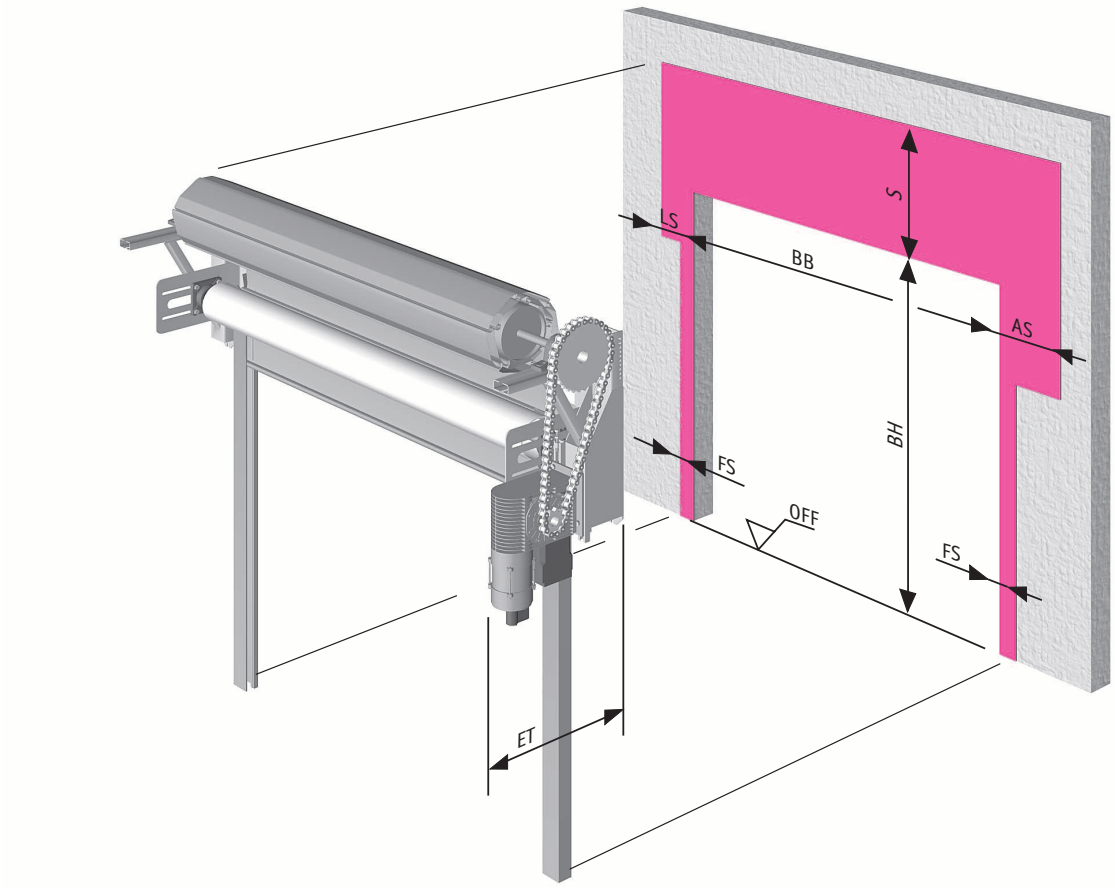


KE = chain (emergency chain drive)



- FS (dimension requirement) see page 8
- Drive (dimension requirement) see page 30

The headroom height S is increased by 210 mm for doors with press shaft.



13. Clearance tubular motor for headroom attachment

Required fitting widths

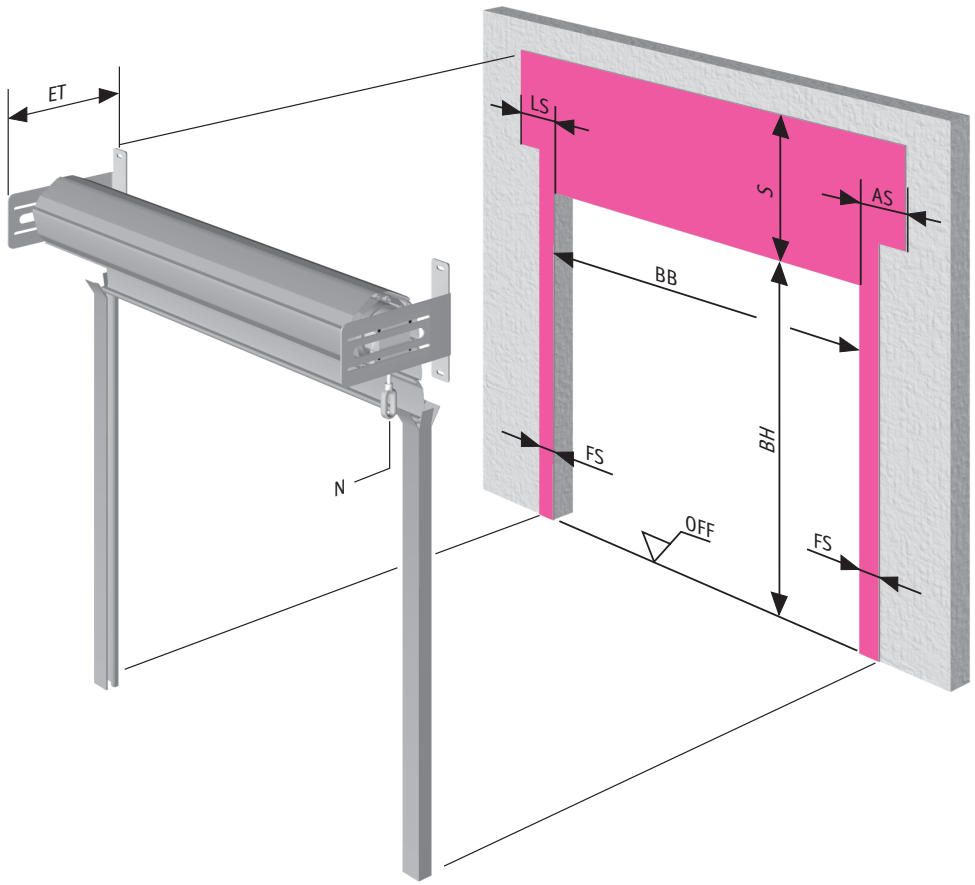
Guide rail type:		FS 60 (ALU)	FS 85	FS 110
FS		60	85	110
LS		140	165	190
AS	TRR 1.14	150	175	200
	TRR 2.18	150	175	200
	TRR 3.23	150	175	200

FS (dimension requirement) see page 8

- Tubular motors only available in 230 V~ (alternating current).
- Only available with 133mm diameter tube.
- Not suitable for high number of door cycles.
- Standard operation is via a key-operated switch; impulse control is possible.
- Max. duty cycle = 5 Min. at an ambient temperature of 20° C.
- Max. number of door cycles per hour = 3x.

All measurements in mm

Mounting on any wall material.



13. Clearance tubular motor for headroom attachment

= Insertion depth (ET)
= Headroom height (S)

Door height in mm	Profile ThermoTeck	Profile 6010 6010 Visio	Rolling grating TG-A/V/S/E	Rolling grating TG-X
4500	574	506	490	526
	494	426	410	446
4250	574	506	490	526
	494	426	410	446
4000	544	498	490	526
	464	418	410	446
3750	540	470	490	526
	460	410	410	446
3500	534	470	465	488
	454	410	410	410
3250	534	470	465	488
	454	410	410	410
3000	508	466	465	470
	428	410	410	410
2750	496	466	445	470
	416	410	410	410
2500	494	466	445	470
	414	410	410	410
2250	494	466	445	458
	414	410	410	410
2000	464	449	445	440
	410	410	410	410

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14. Clearance tubular motor for lateral attachment

Required fitting widths

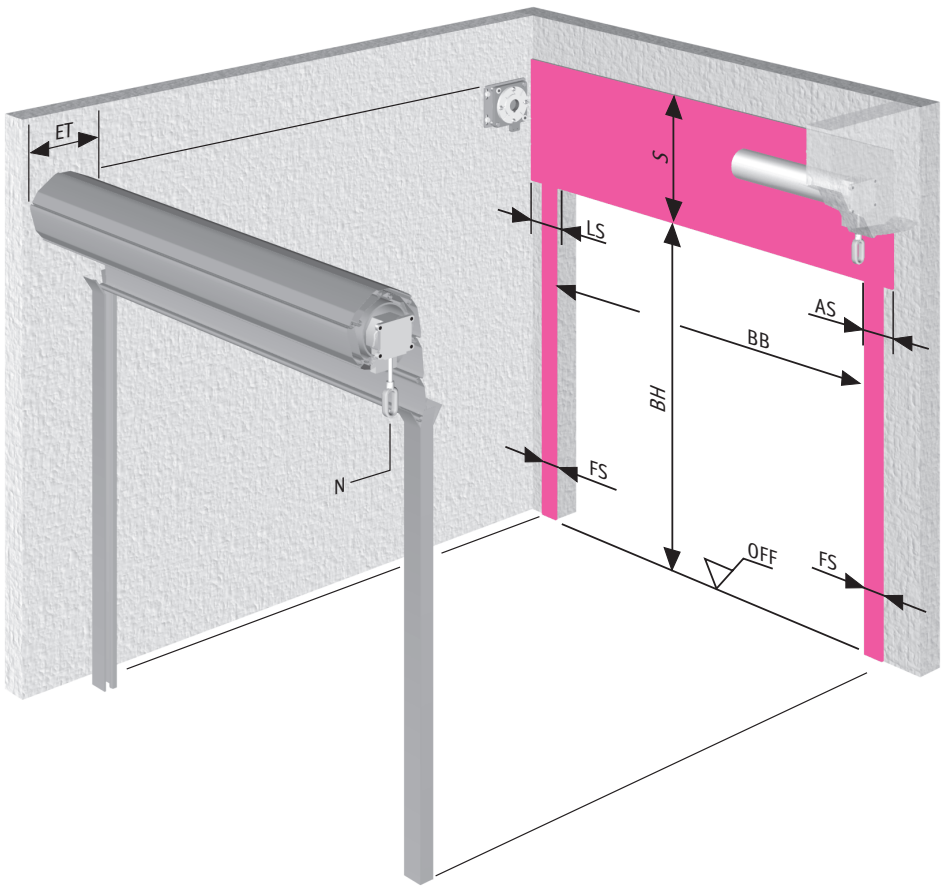
Guide rail type:		FS 60 (ALU)	FS 85	FS 110
FS		60	85	110
LS		125	150	175
AS	TRR 1.14	135	160	185
	TRR 2.18	135	160	185
	TRR 3.23	135	160	185

FS (dimension requirement) see page 8

- Tubular motors only available in 230 V~ (alternating current).
- Only available with 133mm diameter tube.
- Not suitable for high number of door cycles.
- Standard operation is via a key-operated switch; impulse control is possible.
- Max. duty cycle = 5 Min. at an ambient temperature of 20° C.
- Max. number of door cycles per hour = 3x.

All measurements in mm

Mounting on any wall material.



14. Clearance tubular motor for lateral attachment

= Insertion depth (ET)
= Headroom height (S)

Door height in mm	Profile ThermoTeck	Profile 6010 6010 Visio	Rolling grating TG-A/V/S/E	Rolling grating TG-X
4500	574	506	490	526
	494	426	410	446
4250	574	498	490	488
	494	418	410	408
4000	544	498	490	488
	464	418	410	408
3750	540	498	490	488
	460	418	410	408
3500	534	470	460	488
	454	390	380	408
3250	534	470	460	470
	454	390	380	390
3000	508	470	460	470
	428	390	380	390
2750	496	462	420	446
	416	382	340	366
2500	494	462	420	446
	414	382	340	366
2250	494	462	420	446
	414	382	340	366
2000	458	428	420	410
	378	348	340	330

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15. Clearance direct drive with wall flange bearing

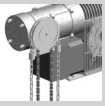
Required fitting widths

Guide rail type:		FS 85	FS 110
FS		85	110
LS / LS1 / LS2		120 / 110 / 200	145 / 135 / 250
AS with KU	TAR 1.14	300	325
	TAR 2.22	300	325
	TAR 3.30	300	325
	TAR 3.42	300	325
	TAR 3.50	300	325
	TAR 5.75	300	325
	TAR 6.100	315	340
AS with KE	TAR 1.14	240	265
with KKU	TAR 2.22	240	265
with NWT	TAR 3.30	240	265
	TAR 3.42	240	265
	TAR 3.50	-	-
	TAR 5.75	-	-
	TAR 6.100	-	-

KU = crank (emergency crank handle)



KE = chain (emergency chain drive)



KKU = universal joint crank (folding)



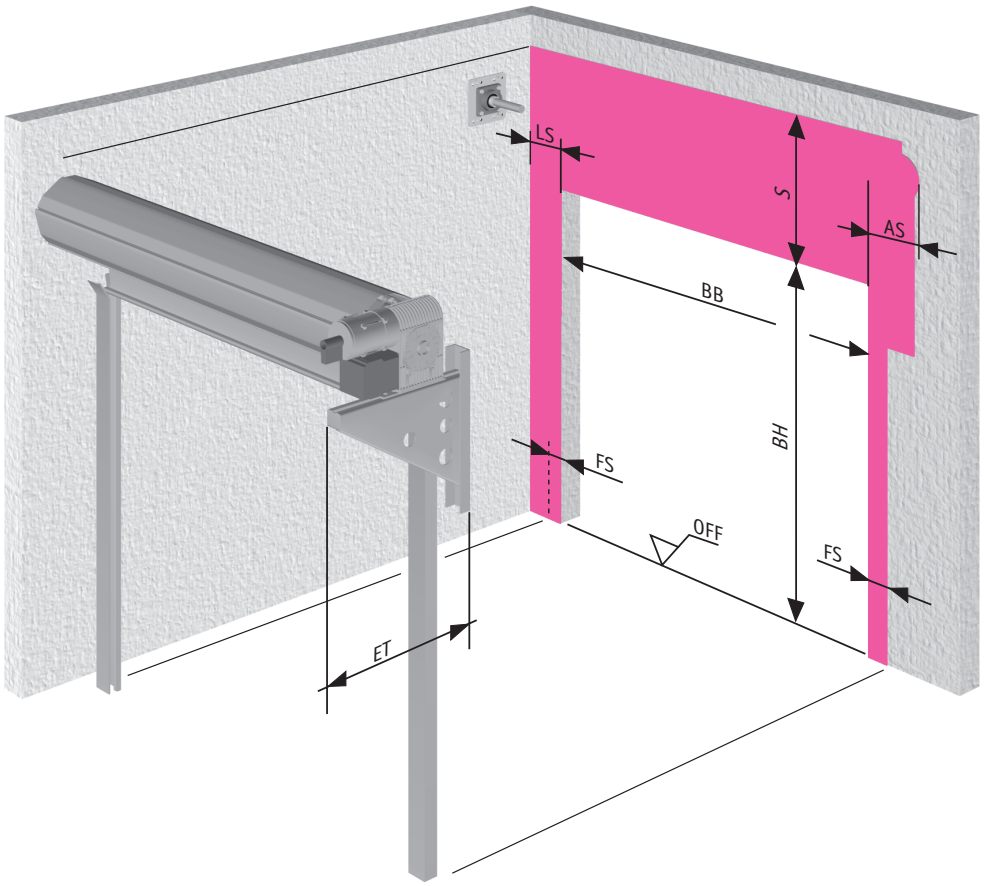
NWT = Emergency operation (with awning rod) only up to TAR 2.22

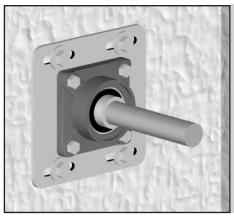


- FS (dimension requirement) see page 8
- Drive (dimension requirement) see page 30

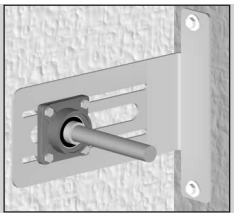
All measurements in mm

Mounting on any wall material.

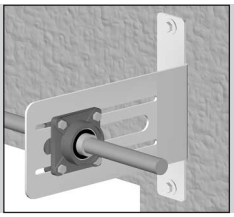




Wall flange bearing simple version (LS) up to TAR 3.50



Wall flange bearing with console, installation on one side (1 shutter LS1) up to TAR 3.42



Wall flange bearing with console, installation on both sides (2 shutters - LS2 / not possible for TAR 5.75 and TAR 6.100) up to 2x TAR 2.22

16. Clearance direct drive for lateral attachment

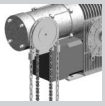
Required fitting widths

Guide rail type:		FS 85	FS 110
FS		85	110
LS		175	200
AS with KU	TAR 1.14	315	340
	TAR 2.22	315	340
	TAR 3.30	315	340
	TAR 3.42	315	340
	TAR 3.50	315	340
	TAR 5.75	315	340
	TAR 6.100	330	355
AS with KE with KKU with NWT	TAR 1.14	315	340
	TAR 2.22	315	340
	TAR 3.30	315	340
	TAR 3.42	315	340
	TAR 3.50	-	-
	TAR 5.75	-	-
	TAR 6.100	-	-

KU = crank (emergency crank handle)



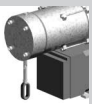
KE = chain (emergency chain drive)



KKU = universal joint crank (folding)



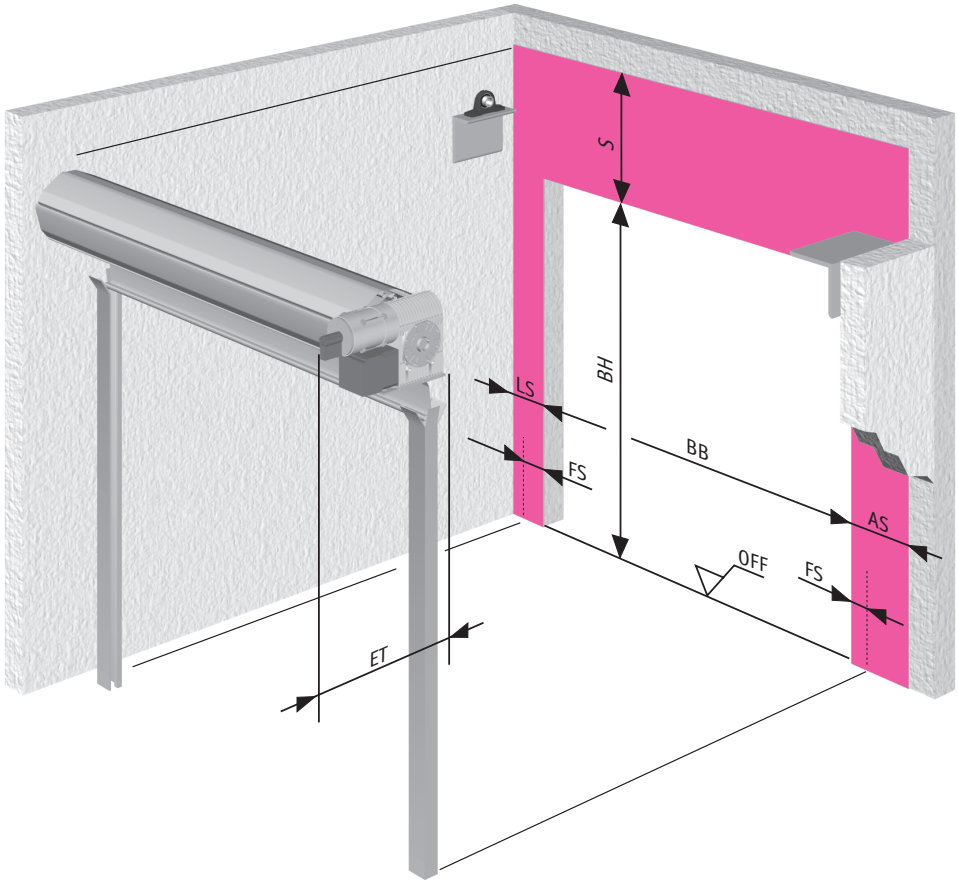
NWT = Emergency operation (with awning rod) only up to TAR 2.22



- FS (dimension requirement) see page 8
- Drive (dimension requirement) see page 30

All measurements in mm

Mounting on any wall material.



17. Clearance direct drive for roof installation

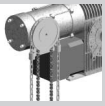
Required fitting widths

Guide rail type:		FS 85	FS 110
FS		85	110
LS		250	275
AS with KU	TAR 1.14	300	325
	TAR 2.22	300	325
	TAR 3.30	300	325
	TAR 3.42	300	325
	TAR 3.50	300	325
	TAR 5.75	300	325
	TAR 6.100	315	340
AS with KE	TAR 1.14	290	315
with KKU	TAR 2.22	290	315
with NWT	TAR 3.30	290	315
	TAR 3.42	290	315
	TAR 3.50	-	-
	TAR 5.75	-	-
	TAR 6.100	-	-

KU = crank (emergency crank handle)



KE = chain (emergency chain drive)



KKU = universal joint crank (folding)



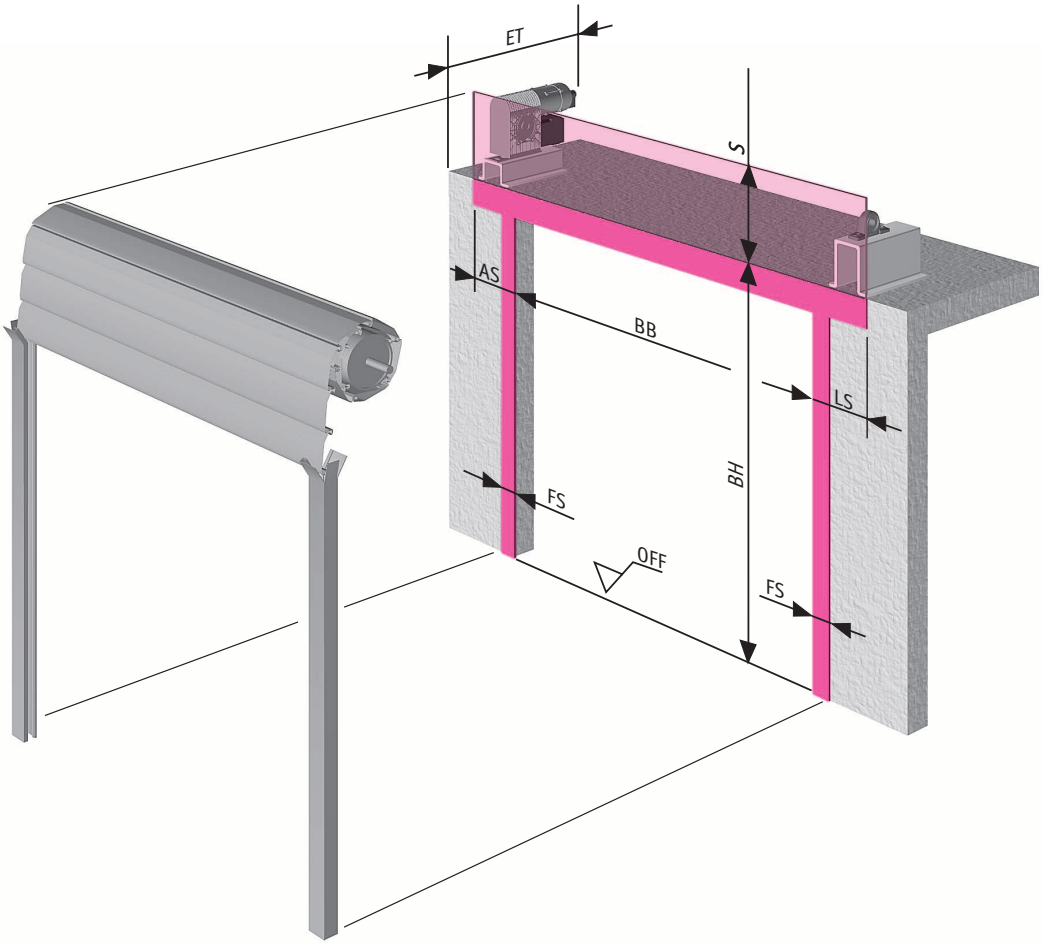
NWT = Emergency operation (with awning rod) only up to TAR 2.22



- FS (dimension requirement) see page 8
- Drive (dimension requirement) see page 30

All measurements in mm

Mounting on any wall material.



18. Clearance direct drive for hanging installation

Required fitting widths

Guide rail type:		FS 85	FS 110
FS		85	110
LS		175	200
AS with KU	TAR 1.14	300	325
	TAR 2.22	300	325
	TAR 3.30	300	325
	TAR 3.42	300	325
	TAR 3.50	300	325
	TAR 5.75	-	-
	TAR 6.100	-	-
AS with KE with KKU	TAR 1.14	240	265
	TAR 2.22	240	265
	TAR 3.30	240	265
	TAR 3.42	240	265
	TAR 3.50	-	-
	TAR 5.75	-	-
	TAR 6.100	-	-

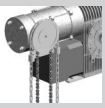
All measurements in mm

Mounting on any wall material.

KU = crank (emergency crank handle)



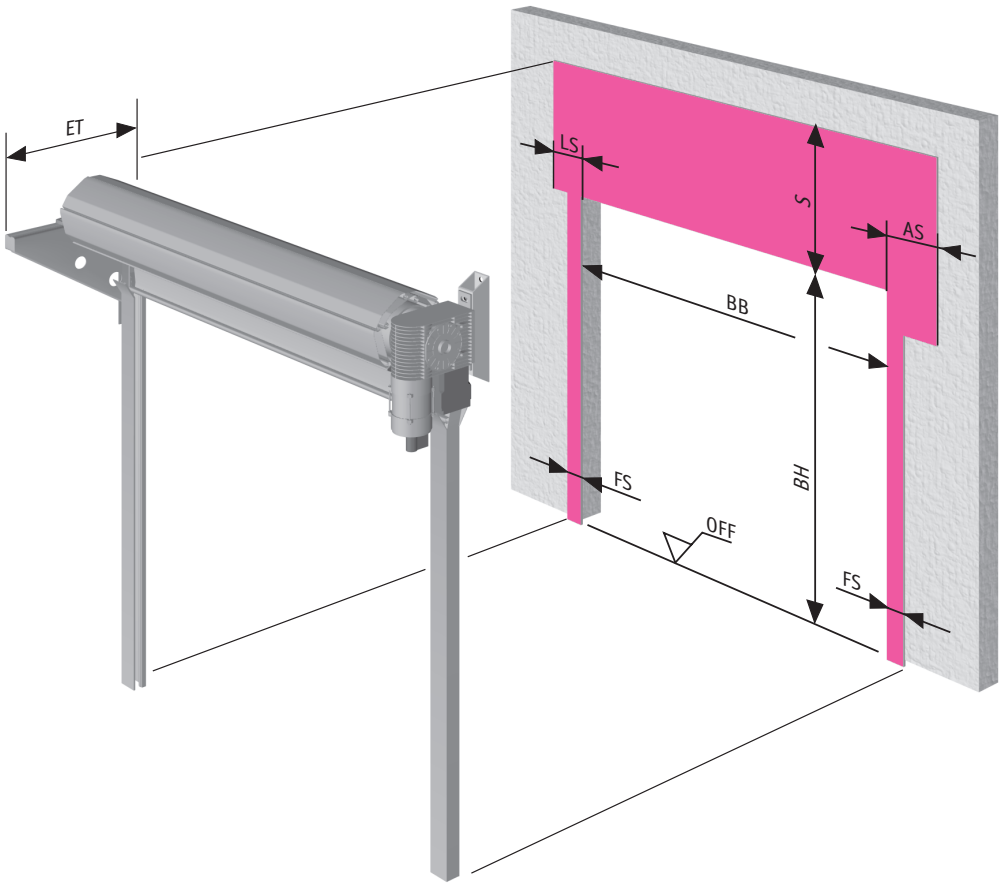
KE = chain (emergency chain drive)



KKU = universal joint crank (folding)



- FS (dimension requirement) see page 8
- Drive (dimension requirement) see page 30



19. Clearance with fixed pass door

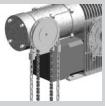
Required fitting widths

Guide rail type:		FS 85	FS 110
FS		85	110
LS		175	200
AS with KU	TAR 1.14	300	325
	TAR 2.22	300	325
	TAR 3.30	300	325
	TAR 3.42	300	325
	TAR 3.50	300	325
	TAR 5.75	300	325
	TAR 6.100	315	340
AS with KE with KKU with NWT	TAR 1.14	240	265
	TAR 2.22	240	265
	TAR 3.30	240	265
	TAR 3.42	240	265
	TAR 3.50	-	-
	TAR 5.75	-	-
	TAR 6.100	-	-

KU = crank (emergency crank handle)



KE = chain (emergency chain drive)



KKU = universal joint crank (folding)



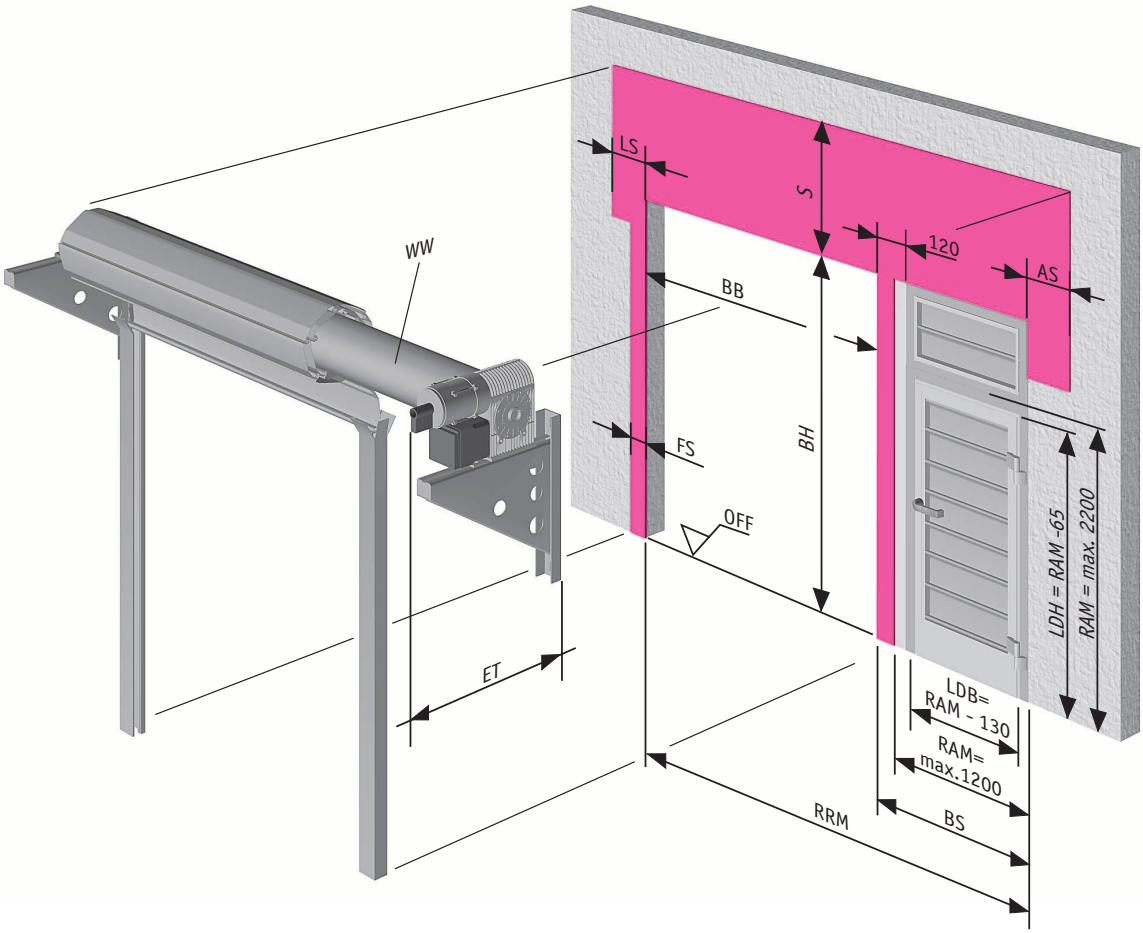
NWT = Emergency operation (with awning rod) only up to TAR 2.22



- FS (dimension requirement) see page 8
- Drive (dimension requirement) see page 30

All measurements in mm

Mounting on any wall material.



Shaft extension (WW): Width side frame - FS

20. Clearance with swing-aside pass door

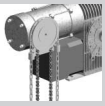
Required fitting widths

Guide rail type:		FS 85	FS 110
FS		85	110
LS		175	200
AS with KU	TAR 1.14	300	325
	TAR 2.22	300	325
	TAR 3.30	300	325
	TAR 3.42	300	325
	TAR 3.50	300	325
	TAR 5.75	300	325
	TAR 6.100	315	340
AS with KE with KKU with NWT	TAR 1.14	240	265
	TAR 2.22	240	265
	TAR 3.30	240	265
	TAR 3.42	240	265
	TAR 3.50	-	-
	TAR 5.75	-	-
	TAR 6.100	-	-

KU = crank (emergency crank handle)



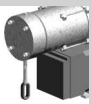
KE = chain (emergency chain drive)



KKU = universal joint crank (folding)



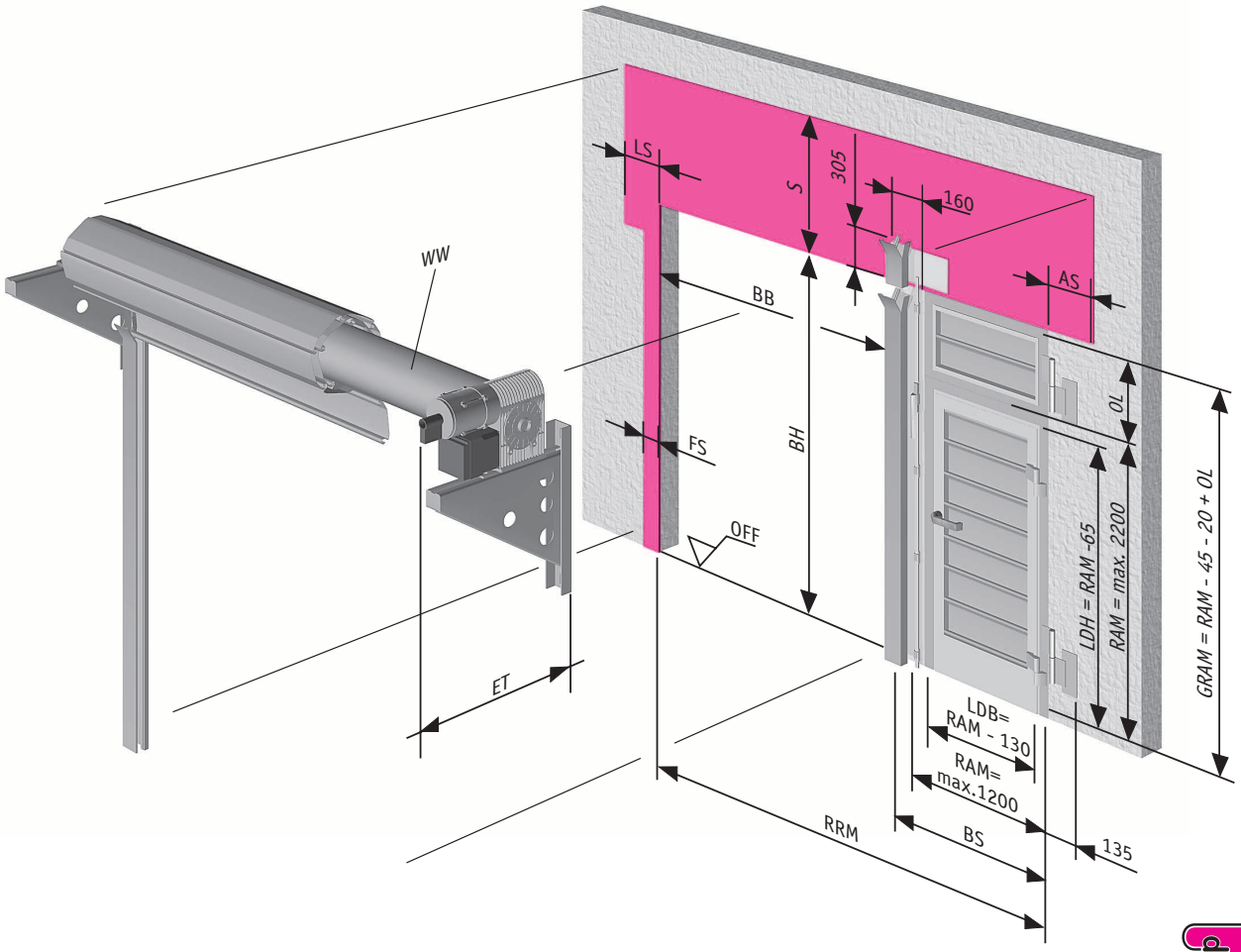
NWT = Emergency operation (with awning rod) only up to TAR 2.22



- FS (dimension requirement) see page 8
- Drive (dimension requirement) see page 30

All measurements in mm

Mounting on any wall material.



Shaft extension (WW): Width side frame - FS
Width side frame: RAM + 160 + 50
Order measurement - door width: RRM - RAM - 180

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21. Clearance direct drive for ceiling installation

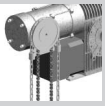
Required fitting widths

Guide rail type:		FS 85	FS 110
FS		85	110
LS		175	200
AS with KU	TAR 1.14	330	355
	TAR 2.22	330	355
	TAR 3.30	330	355
	TAR 3.42	330	355
	TAR 3.50	330	355
	TAR 5.75	-	-
	TAR 6.100	-	-
AS with KE	TAR 1.14	330	355
with KKU	TAR 2.22	330	355
with NWT	TAR 3.30	330	355
	TAR 3.42	330	355
	TAR 3.50	-	-
	TAR 5.75	-	-
	TAR 6.100	-	-

KU = crank (emergency crank handle)



KE = chain (emergency chain drive)



KKU = universal joint crank (folding)



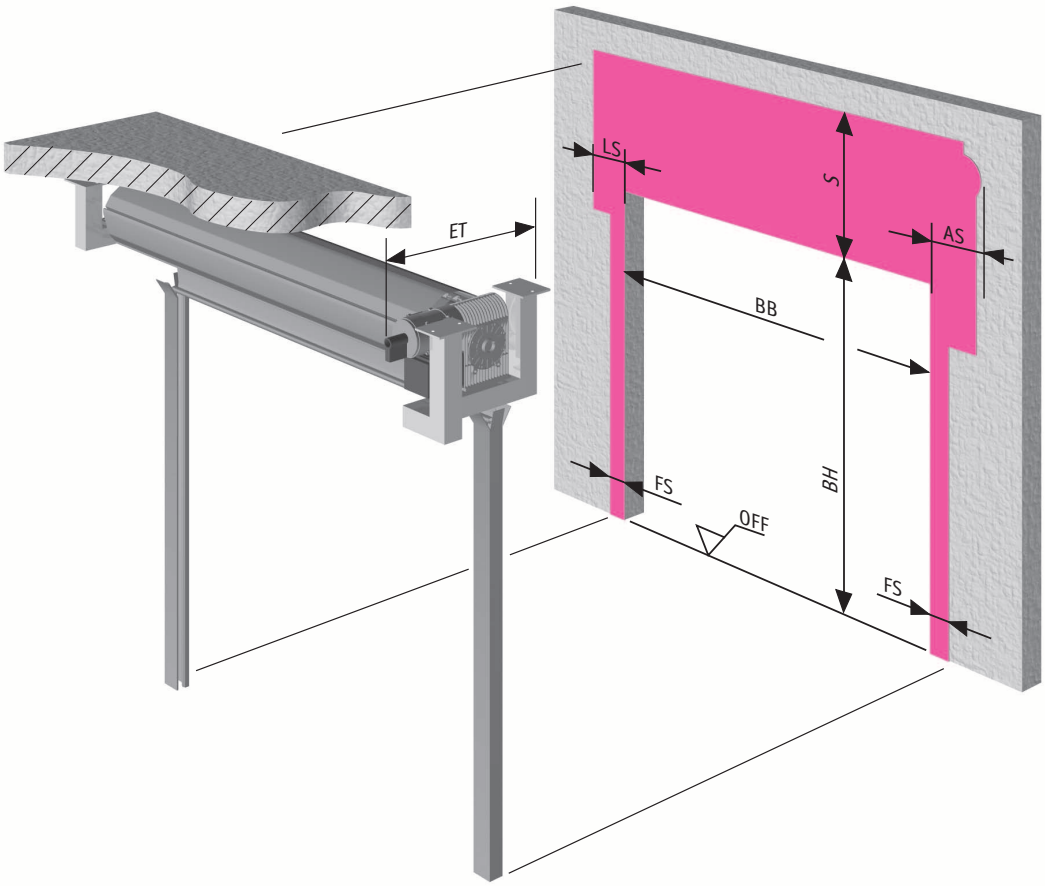
NWT = Emergency operation (with awning rod) only up to TAR 2.22



- FS (dimension requirement) see page 8
- Drive (dimension requirement) see page 30

All measurements in mm

Installation only on steel or concrete.



22. Clearance direct drive with external operation

Required fitting widths

Guide rail type:		FS 85	FS 110
FS		85	110
LS		175	200
AS with KU	TAR 1.14	-	-
	TAR 2.22	300	325
	TAR 3.30	300	325
	TAR 3.42	300	325
	TAR 3.50	-	-
	TAR 5.75	-	-
	TAR 6.100	-	-

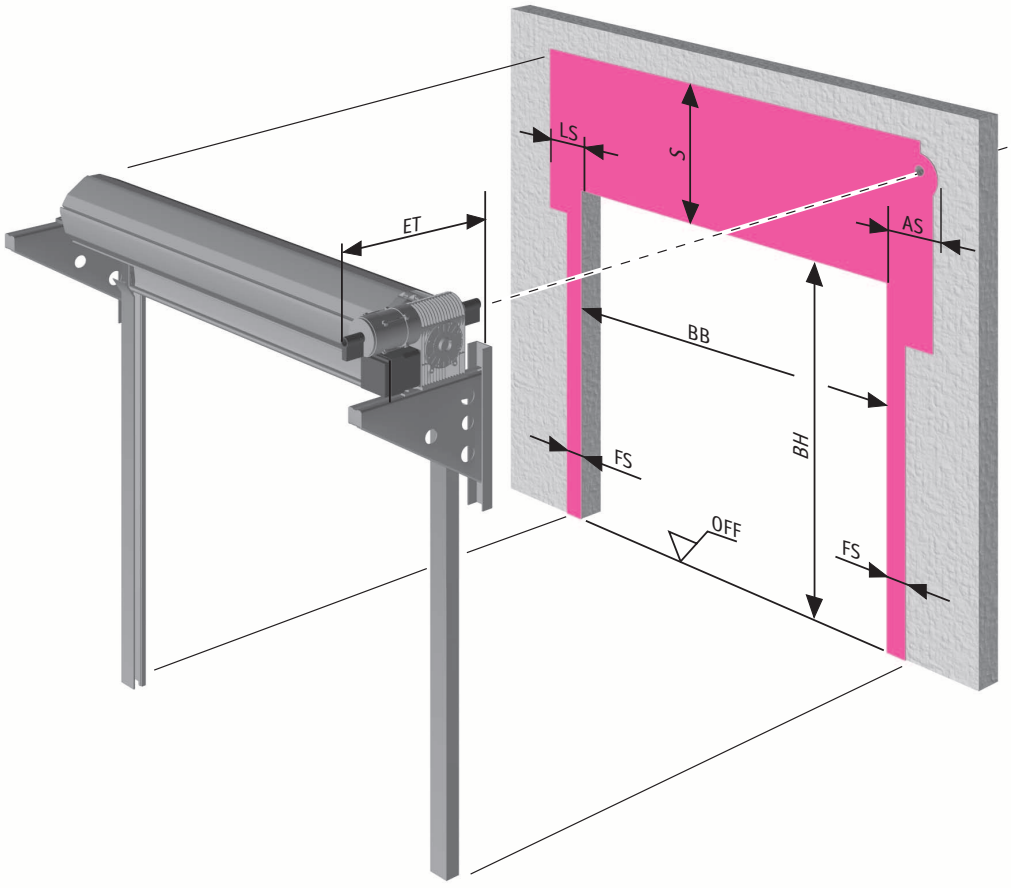
KU = crank (emergency crank handle)



- FS (dimension requirement) see page 8
- Drive (dimension requirement) see page 30

All measurements in mm

Mounting on any wall material.



Crank 400 mm extended!

23. Weld-on plates for consoles

Starting points for weld-on plates (lower edge headroom to lower edge plate) for direct drives TAR 1.14 to TAR 3.500

Door height in mm:	**Rolling grating** TG-S, TG-A, TG-V, TG-E, TG-X		**6010** Alu/Steel Visio		**4020** Alu		**ThermoTeck** Steel	
	Drive X1	Bearing X2	Drive X1	Bearing X2	Drive X1	Bearing X2	Drive X1	Bearing X2
8000	183	81	163	60	120	18	150	48
7500	185	83	170	68	120	18	150	48
7000	193	91	170	68	140	38	155	53
6500	200	98	173	70	140	38	165	63
6000	210	108	180	78	140	38	165	63
5500	243	141	230	128	198	96	213	111
5000	253	151	235	133	198	96	215	113
4500	263	161	245	143	206	104	220	118
4000	270	168	245	143	222	120	234	132
3500	273	171	255	153	222	120	234	132
3000	285	183	260	158	245	143	255	153
2500	295	193	280	178	248	146	255	153
2000	303	201	285	183	256	154	276	174

Weld-on plate dimensions: 500 x 200 x 6 mm (size 1)

All measurements in mm

Starting points for weld-on plates (lower edge headroom to lower edge plate) for direct drives TAR 5.75, 6.100 and chain drive up to TKR 5.40

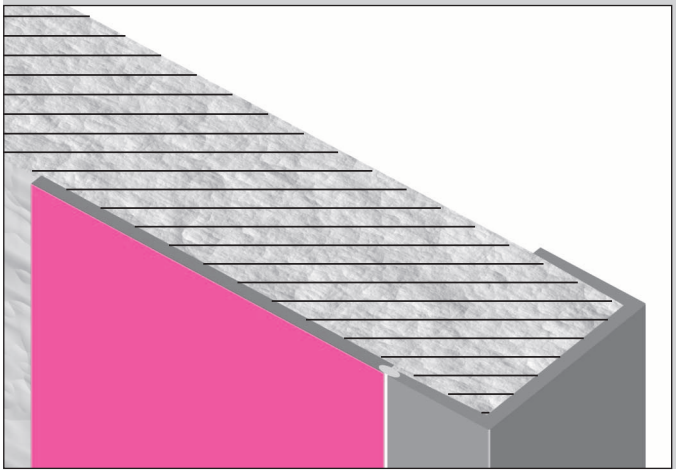
Door height in mm:	**Rolling grating** TG-S, TG-A, TG-V, TG-E, TG-X		**6010** Alu/Steel Visio		**4020** Alu		**ThermoTeck** Steel	
	Drive X1	Bearing X2	Drive X1	Bearing X2	Drive X1	Bearing X2	Drive X1	Bearing X2
8000	432	415	412	395	369	352	399	382
7500	434	417	419	402	369	352	399	382
7000	442	425	419	402	389	372	404	387
6500	449	432	422	405	389	372	414	397
6000	459	442	429	412	389	372	414	397
5500	492	475	479	462	447	425	462	445
5000	502	485	484	467	447	425	464	447
4500	512	495	494	477	455	438	469	452
4000	519	502	494	477	471	454	483	466
3500	524	507	504	487	471	454	483	466
3000	534	517	509	492	494	477	504	487
2500	544	527	529	512	497	480	504	487
2000	552	535	534	517	505	488	525	508

Weld-on plate dimensions: 900 x 200 x 8 mm (size 2)

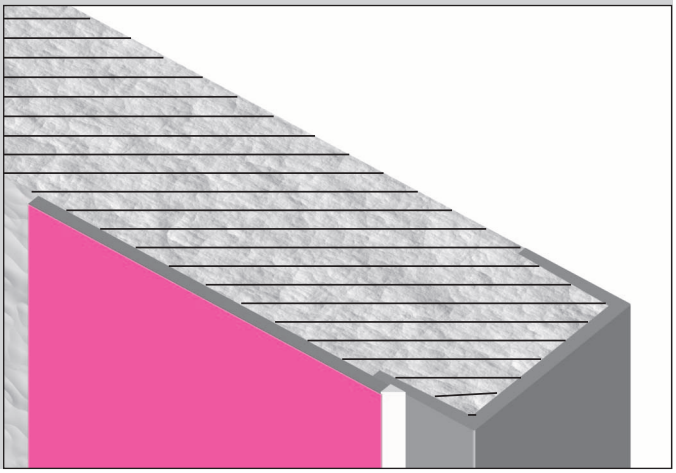
All measurements in mm

23. Weld-on plates for consoles

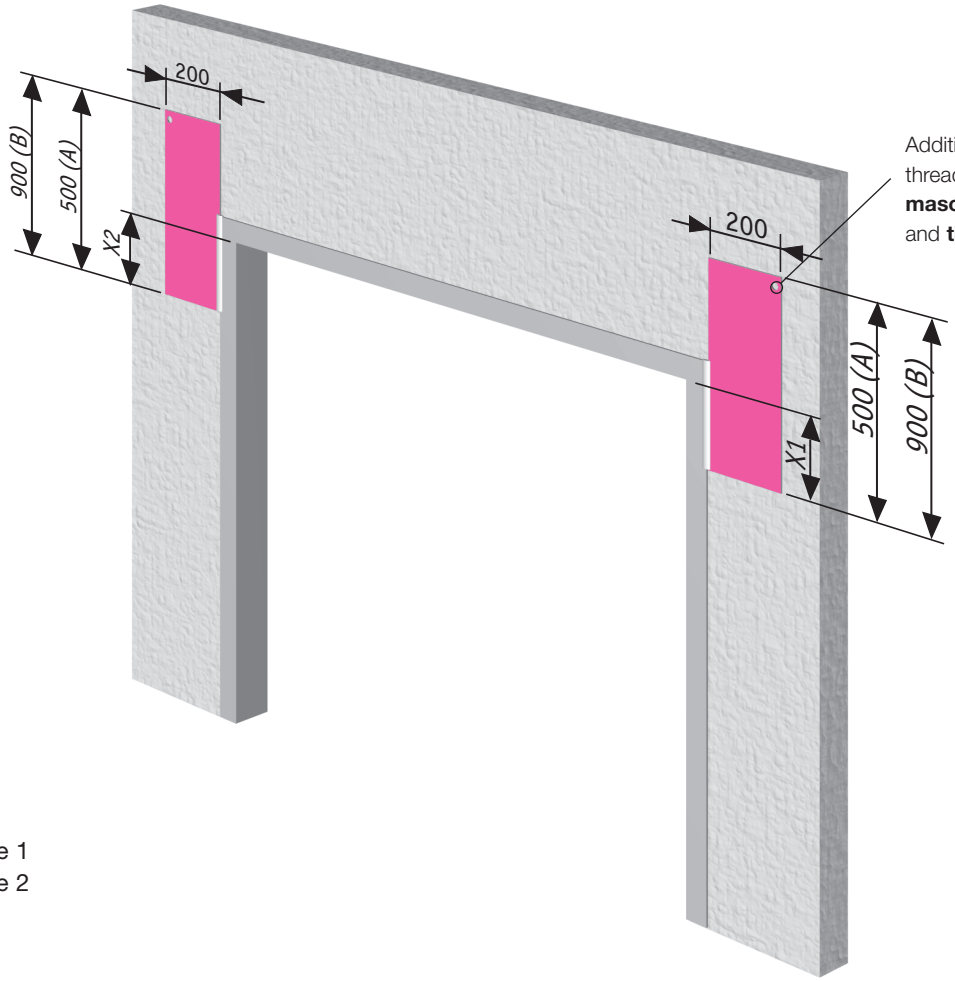
Version 1
Welding plate against profile.
Steel profile < FS-width



Version 2
Welding plate against profile.
Steel profile > FS-width



Advice:
- Weld-on plates subject to additional charge.



Additional attachment via threaded rods required for **masonry, aerated concrete** and **timber**.

A: Size 1
B: Size 2

24. 3-D view shutter cladding mounted with plates

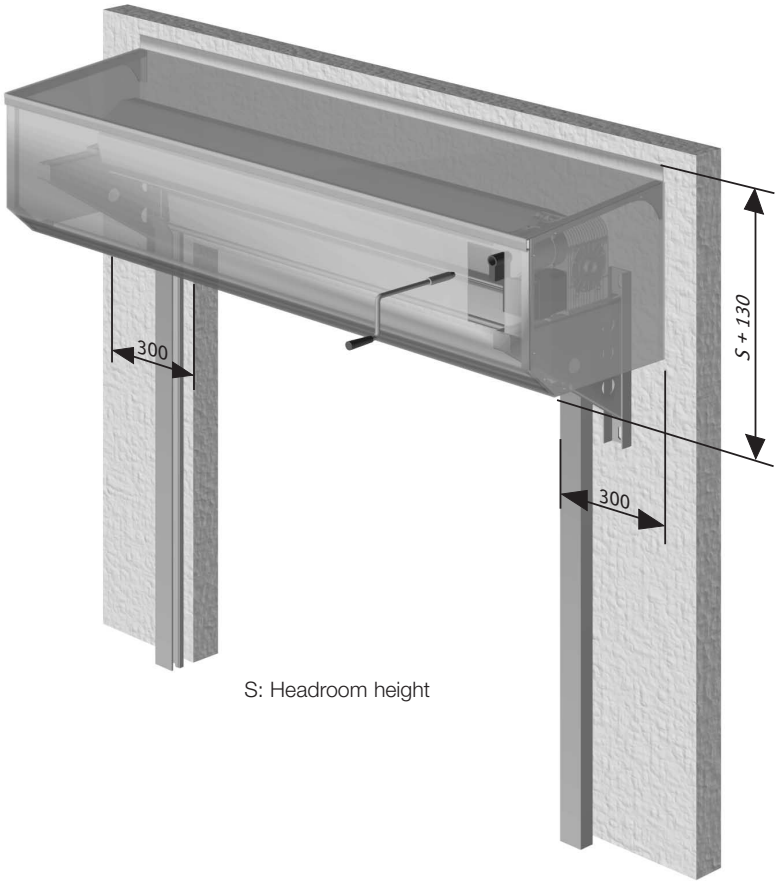
The max. Plate length for the upper and front plate is 3.000 mm.

For outside shutters we recommend drives with IP65 (additional charge).
Roller box dimensions, special designs and special sizes available on request.

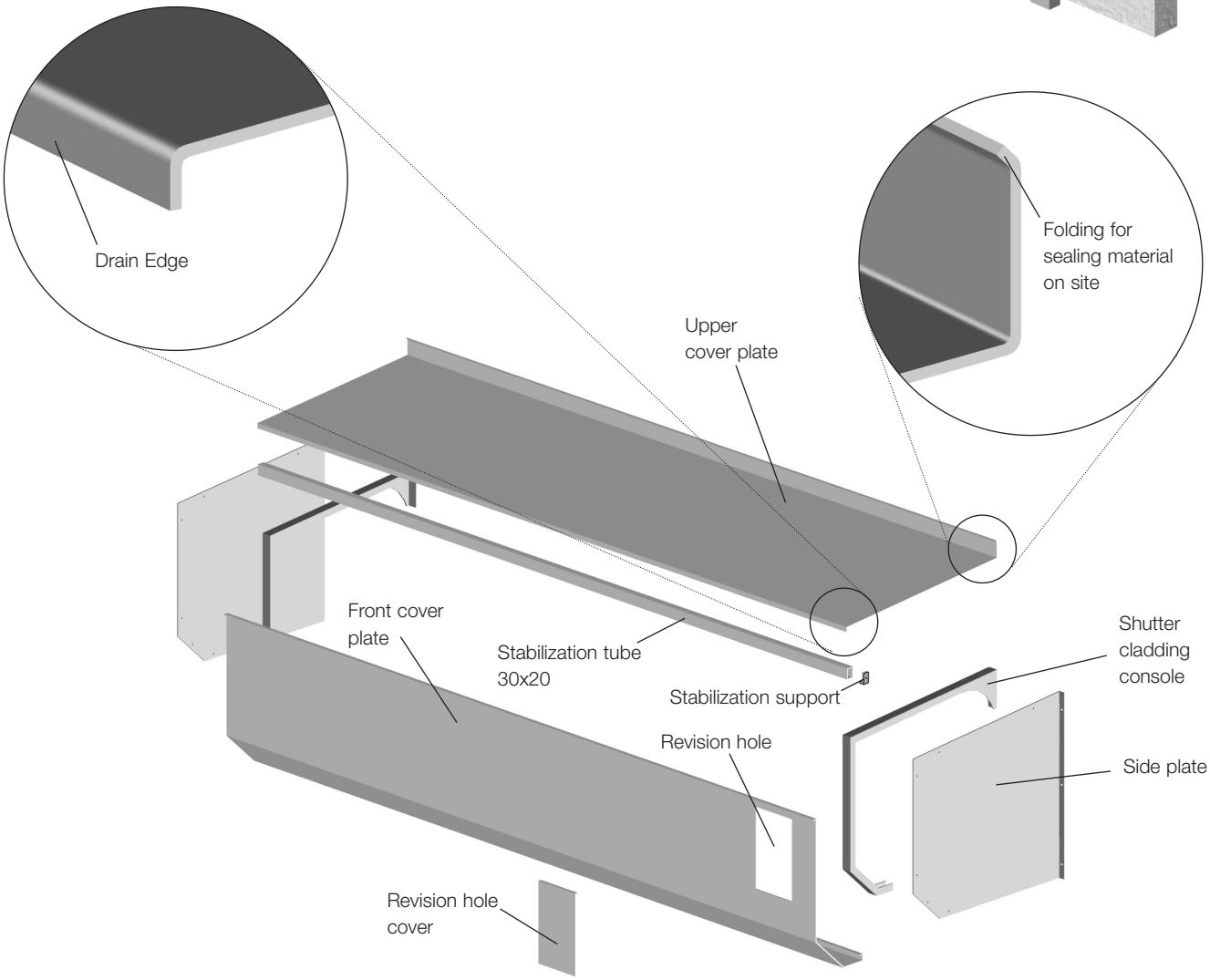
Important:
For chain drives the panelling only covers the rolling shutter cladding.
The chain drive itself has to be covered on site.

Available:
- in sizes 0, 1, 2 and 3,
- in aluminium or steel.

Advice:
The emergency crank handle is extended by 400 mm!



S: Headroom height



25. Drive data

Direct drives

Name of operator	TAR	1.14	2.22	3.30	3.42	3.50	5.75	6.100
Output torque	Nm	140	220	300	420	500	750	1.000
Output speed	min^{-1}	12	12	12	12	12	10	9
Motor output	KW	0,37	0,75	1,1	1,1	1,1	1,4	1,5
Operating voltage	V	400 / 3~	400 / 3~	400 / 3~	400 / 3~	400 / 3~	400 / 3~	400 / 3~
Mains frequency	Hz	50	50	50	50	50	50	50
Motor rated current	A	2,1	2,8	3,6	4,3	4,5	5,8	6,1
Motor duty cycle	% ED	60	60	60	60	60	60	60
Feed cable provided by the customer	mm^2	5 x 1,5	5 x 1,5	5 x 1,5	5 x 1,5	5 x 1,5	5 x 1,5	5 x 1,5
Fuse protection provided by the customer	A	10	10	10	10	10	10	10
Degree of protection	IP	54	54	54	54	54	54	54
Temperature range	°C	-20 / +60	-20 / +60	-20 / +60	-20 / +60	-20 / +60	-20 / +60	-20 / +60
Continuous sound intensity level	db (A)	< 70	< 70	< 70	< 70	< 70	< 70	< 70
Weight per unit	kg	25	20	23	27	29	33	67
Hollow tube diameter	mm	30	30	30	40	40	50	50

Chain drives

Name of operator	TKR	1.10	2.15	3.22	4.30	5.40	6.60	7.97
Output torque	Nm	100	150	220	300	400	600	970
Output speed	min^{-1}	24	24	24	24	24	24	24
Motor output	KW	0,55	0,75	1,1	1,5	2,2	3,0	4,0
Operating voltage	V	400 / 3~	400 / 3~	400 / 3~	400 / 3~	400 / 3~	400 / 3~	400 / 3~
Mains frequency	Hz	50	50	50	50	50	50	50
Motor rated current	A	1,9	2,4	2,75	4,2	6,5	7,8	10
Motor duty cycle	% ED	40	40	40	40	40	100	100
Feed cable provided by the customer	mm^2	5 x 1,5	5 x 1,5	5 x 1,5	5 x 1,5	5 x 1,5	5 x 1,5	5 x 1,5
Fuse protection provided by the customer	A	10	10	10	10	10	10	10
Degree of protection	IP	54	54	54	54	54	54	54
Temperature range	°C	-20 / +60	-20 / +60	-20 / +60	-20 / +60	-20 / +60	-20 / +60	-20 / +60
Continuous sound intensity level	db (A)	< 70	< 70	< 70	< 70	< 70	< 70	< 70
Weight per unit	kg	20	21	22	31	33	85	92
Hollow tube diameter	mm	30	30	30	40	40	50	50
Chain drive reduction		3:1	3:1	3:1	3:1	4:1	4:1	4:1
Number of teeth bottom	Piece	14	12	10	10	10	12	12
Number of teeth top	Piece	42	36	30	30	40	45	45
Number of elements	Piece	114	92	64	66	74	82	82
Chain division	Customs	1/2 x 5/16	5/8 x 3/8	1 x 17,02	1 x 17,02	1 x 17,02	1 1/4	1 1/2

Tubular motors

Name of operator	TRR	1.14	2.18	3.23
Output torque	Nm	140	180	230
Output speed	min^{-1}	12	12	12
Motor output	KW	0,7	0,75	0,87
Operating voltage	V	230 / 1~	230 / 1~	230 / 1~
Mains frequency	Hz	50	50	50
Motor rated current	A	3,0	3,4	3,8
Motor duty cycle	Min.	5	5	5
Feed cable provided by the customer	mm^2	3 x 1,5	3 x 1,5	3 x 1,5
Fuse protection provided by the customer	A	10	10	10
Degree of protection	IP	44	44	44
Temperature range	°C	-20 / +50	-20 / +50	-20 / +50
Continuous sound intensity level	db (A)	< 70	< 70	< 70
Weight per unit	kg	14	14	14



- Max. duty cycle = 5 Min. at an ambient temperature of 20° C.
- Max. number of door cycles per hour = 3x.

26. Table of reference values ThermoTeck steel, galvanized

S = Min. headroom
ET = Installation depth with emergency crank handle
UFU = Underfill height
T = Opening time of the shutter system approx. in seconds

Height		Shaft diameter 159 mm													177 mm			219 mm	
	S	680	680	680	680	680	680	680	680	680	680	680	680	680	710	710	710	740	740
	ET	823	898	898	898	911	911	911	911	911	911	911	911	925	940	940	940	1040	1040
8000	UFU	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	80	80
	T	37	37	37	37	37	37	37	37	37	37	37	37	37	34	34	34	31	31
	S	680	680	680	680	680	680	680	680	680	680	680	680	680	680	710	710	710	735
	ET	823	898	898	898	911	911	911	911	911	911	911	911	911	925	940	940	940	1038
7750	UFU	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	80
	T	36	36	36	36	36	36	36	36	36	36	36	36	36	36	33	33	33	31
	S	680	680	680	680	680	680	680	680	680	680	680	680	680	680	694	694	694	735
	ET	823	898	898	898	898	911	911	911	911	911	911	911	911	911	932	932	932	1038
7500	UFU	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	80
	T	35	35	35	35	35	35	35	35	35	35	35	35	35	35	33	33	33	30
	S	680	680	680	680	680	680	680	680	680	680	680	680	680	680	678	678	678	730
	ET	823	898	898	898	898	911	911	911	911	911	911	911	911	911	924	924	924	1035
7250	UFU	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	80
	T	33	33	33	33	33	33	33	33	33	33	33	33	33	33	33	33	33	29
	S	670	670	670	670	670	670	670	670	670	670	670	670	670	670	678	678	678	730
	ET	818	818	893	893	893	893	906	906	906	906	906	906	906	906	924	924	924	1035
7000	UFU	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	80
	T	33	33	33	33	33	33	33	33	33	33	33	33	33	33	31	31	31	28
	S	650	650	650	650	650	650	650	650	650	650	650	650	650	650	670	670	670	710
	ET	808	808	883	883	883	883	896	896	896	896	896	896	896	896	920	920	920	940
6750	UFU	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60
	T	33	33	33	33	33	33	33	33	33	33	33	33	33	33	31	31	31	27
	S	650	650	650	650	650	650	650	650	650	650	650	650	650	650	668	668	668	668
	ET	808	808	883	883	883	883	883	896	896	896	896	896	896	896	919	919	919	919
6500	UFU	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60
	T	31	31	31	31	31	31	31	31	31	31	31	31	31	31	30	30	30	30
	S	650	650	650	650	650	650	650	650	650	650	650	650	650	650	650	668	668	668
	ET	808	808	808	883	883	883	883	896	896	896	896	896	896	896	896	919	919	919
6250	UFU	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60
	T	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	28	28	28
	S	650	650	650	650	650	650	650	650	650	650	650	650	650	650	650	648	648	648
	ET	808	808	808	883	883	883	883	883	896	896	896	896	896	896	896	895	909	909
6000	UFU	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60
	T	29	29	29	29	29	29	29	29	29	29	29	29	29	29	29	28	28	28
	S	644	644	644	644	644	644	644	644	644	644	644	644	644	644	644	634	634	634
	ET	805	805	805	805	880	880	880	880	893	893	893	893	893	893	893	888	888	902
5750	UFU	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60
	T	28	28	28	28	28	28	28	28	28	28	28	28	28	28	28	28	28	28
	S	584	584	584	584	584	584	584	584	584	584	584	584	584	584	584	598	598	598
	ET	790	790	790	790	865	865	865	865	865	878	878	878	878	878	878	885	885	885
5500	UFU	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	60	60	60
	T	28	28	28	28	28	28	28	28	28	28	28	28	28	28	28	27	27	27
	S	580	580	580	580	580	580	580	580	580	580	580	580	580	580	580	580	596	596
	ET	788	788	788	788	788	863	863	863	863	863	876	876	876	876	876	876	884	884
5250	UFU	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	60	60
	T	27	27	27	27	27	27	27	27	27	27	27	27	27	27	27	27	25	25
	S	580	580	580	580	580	580	580	580	580	580	580	580	580	580	580	580	596	596
	ET	788	788	788	788	788	788	863	863	863	863	863	876	876	876	876	876	884	884
5000	UFU	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	60	60
	T	26	26	26	26	26	26	26	26	26	26	26	26	26	26	26	26	24	24
	S	580	580	580	580	580	580	580	580	580	580	580	580	580	580	580	580	574	574
	ET	787	788	788	788	788	788	863	863	863	863	863	863	876	876	876	876	873	873
4750	UFU	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30
	T	24	24	24	24	24	24	24	24	24	24	24	24	24	24	24	24	24	24
	S	570	570	570	570	570	570	570	570	570	570	570	570	570	570	570	570	570	562
	ET	782	783	783	783	783	783	783	858	858	858	858	858	871	871	871	871	871	867
4500	UFU	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30
	T	23	23	23	23	23	23	23	23	23	23	23	23	23	23	23	23	23	23
	S	542	542	542	542	542	542	542	542	542	542	542	542	542	542	542	542	542	558
	ET	768	768	769	769	769	769	769	769	844	844	844	844	844	844	857	857	857	865
4250	UFU	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	30
	T	23	23	23	23	23	23	23	23	23	23	23	23	23	23	23	23	23	22
	S	542	542	542	542	542	542	542	542	542	542	542	542	542	542	542	542	542	554
	ET	768	768	769	769	769	769	769	769	769	844	844	844	844	844	844	857	857	863
4000	UFU	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	30
	T	22	22	22	22	22	22	22	22	22	22	22	22	22	22	22	22	22	21
	S	542	542	542	542	542	542	542	542	542	542	542	542	542	542	542	542	542	542
	ET	768	768	768	769	769	769	769	769	769	844	844	844	844	844	844	844	857	857
3750	UFU	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without
	T	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20
	S	542	542	542	542	542	542	542	542	542	542	542	542	542	542	542	542	542	542
	ET	768	768	768	769	769	769	769	769	769	769	769	844	844	844	844	844	844	844
3500	UFU	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without
	T	19	19	19	19	19	19	19	19	19	19	19	19	19	19	19	19	19	19
	S	506	506	506	506	502	502	502	502	502	502	502	502	502	502	502	502	502	502
	ET	748	748	748	748	749	749	749	749	749	749	749	749	824	824	824	824	824	824
3250	UFU	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without
	T	19	19	19	19	19	19	19	19	19	19	19	19	19	19	19	19	19	19
	S	505	505	505	505	505	500	500	500	500	500	500	500	500	500	500	500	500	500
	ET	747	747	747	747	747	748	748	748	748	748	748	748	748	748	823	823	823	823
3000	UFU	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without
	T	17	17	17	17	17	17	17	17	17	17	17	17	17	17	17	17	17	17
	S	505	505	505	505	505	505	500	500	500	500	500	500	500	500	500	500	500	500
	ET	747	747	747	747	747	747	748	748	748	748	748	748	748	748	748	823	823	823
2750	UFU	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without
	T	16	16	16	16	16	16	16	16	16	16	16	16	16	16	16	16	16	16
	S	505	505	505	505	505	505	505	500	500	500	500	500	500	500	500	500	500	500
	ET	747	747	747	747	747	747	747	748	748	748	748	748	748	748	748	748	748	748
2500	UFU	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without
	T	14	14	14	14	14	14	14	14	14	14	14	14	14	14	14	14	14	14
	S	485	485	485	485	485	485	485	485	485	460	460	460	460	460	460	460	460	460
	ET	727	727	727	727	727	727	727	727	727	728	728	728	728	728	728	728	728	728
2250	UFU	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without
	T	14	14	14	14	14	14	14	14	14	14	14	14	14	14	14	14	14	14
	S	484	484	484	484	484	484	484	484	484	484	484	458	458	458	458	458	458	458
	ET	726	726	726	726	726	726	726	726	726	726	726	727	727	727	727	727	727	727
2000	UFU	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without
	T	13	13	13	13	13	13	13	13	13	13	13	13	13	13	13	13	13	13
Width >		2000		2500		3000		3500		4000		4500		5000		5500		6000	

Shutters for indoors, resistance to wind load class 2 in accordance with DIN EN 13241-1

TAP 3.50
TAP 3.42
TAP 3.30
TAP 2.22
TAP 1.14

26. Table of reference values ThermoTeck steel, galvanized

Advice: The tables of standard values are only for standard shutters.

219 mm				273 mm							298 mm				323 mm				Height
740	740	740	740	754	754	976	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	
1040	1040	1040	1040	664	664	676	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	
80	80	80	80	80	80	80	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	8000
31	31	31	31	29	29	28	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	
735	735	735	735	726	726	950	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	
1038	1038	1038	1038	636	636	650	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	7750
80	80	80	80	80	80	80	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	
31	31	31	31	29	29	28	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	
735	735	735	735	734	734	934	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	
1038	1038	1038	1038	644	644	634	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	7250
80	80	80	80	80	80	80	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	
30	30	30	30	28	28	28	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	
730	730	730	730	730	734	944	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	
1035	1035	1035	1035	1035	644	644	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	7250
80	80	80	80	80	80	80	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	
29	29	29	29	29	27	27	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	
730	730	730	730	730	732	944	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	
1035	1035	1035	1035	1035	642	644	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	7000
80	80	80	80	80	80	80	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	
28	28	28	28	28	26	26	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	
710	710	710	710	710	730	942	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	
1025	1025	1025	1025	1025	640	642	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	6750
60	60	60	60	60	80	80	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	
27	27	27	27	27	25	25	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	
710	710	710	710	710	722	934	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	
940	1025	1025	1025	1025	632	634	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	6500
60	60	60	60	60	80	80	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	
26	26	26	26	26	24	24	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	
691	691	691	691	691	691	924	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	
931	931	1016	1016	1016	1016	624	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	6250
60	60	60	60	60	60	80	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	
26	26	26	26	26	26	24	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	- / -	
691	691	691	691	691	691	902	902	902	902	902	902	902	924	924	924	924	924	952	
931	931	931	1016	1016	1016	602	602	602	602	602	602	602	624	624	624	624	624	652	6000
60	60	60	60	60	60	60	60	60	60	60	60	60	80	80	80	80	80	80	
25	25	25	25	25	25	23	23	23	23	23	23	23	22	22	22	22	22	21	
634	691	691	691	691	691	902	902	902	902	902	902	902	902	924	924	924	924	924	
902	931	931	931	1016	1016	602	602	602	602	602	602	602	602	624	624	624	624	624	5750
60	60	60	60	60	60	60	60	60	60	60	60	60	60	80	80	80	80	80	
28	24	24	24	24	24	22	22	22	22	22	22	22	22	21	21	21	21	20	
598	661	661	661	661	661	872	872	872	872	872	872	872	872	890	890	890	890	890	
899	931	931	931	931	1016	602	602	602	602	602	602	602	602	620	620	620	620	620	5500
60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	
27	23	23	23	23	23	21	21	21	21	21	21	21	21	20	20	20	20	20	
596	601	601	601	601	601	870	870	870	870	870	870	870	870	870	890	890	890	890	
884	901	901	901	901	901	1015	600	600	600	600	600	600	600	600	620	620	620	620	5250
60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	
25	24	24	24	24	24	20	20	20	20	20	20	20	20	20	19	19	19	19	
596	599	599	599	599	599	809	866	866	866	866	866	866	866	866	870	870	870	870	
884	900	900	900	900	900	900	1013	596	596	596	596	596	596	596	600	600	600	600	5000
60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	
24	23	23	23	23	23	23	19	19	19	19	19	19	19	19	19	19	19	19	
574	574	599	599	599	599	809	848	848	848	848	848	848	848	848	850	850	850	850	
873	873	900	900	900	900	900	1004	1004	578	578	578	578	578	578	580	580	580	580	4750
30	30	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	
24	24	21	21	21	21	21	19	19	19	19	19	19	19	19	18	18	18	18	
562	562	599	599	599	599	809	834	834	834	834	834	834	834	834	834	850	850	850	
867	867	900	900	900	900	900	997	997	997	997	575	575	575	575	575	580	580	580	4500
30	30	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	
23	23	20	20	20	20	20	18	18	18	18	18	18	18	18	18	17	17	17	
558	558	558	597	597	597	809	809	834	834	834	834	834	834	834	834	846	846	846	
865	865	865	899	899	899	900	900	997	997	997	997	997	575	575	575	576	576	576	4250
30	30	30	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	
22	22	22	19	19	19	19	19	17	17	17	17	17	17	17	17	16	16	16	
554	554	554	564	564	564	789	807	834	834	834	834	834	834	834	834	834	846	846	
863	863	863	879	879	879	890	899	997	997	997	997	997	997	997	575	575	576	576	4000
30	30	30	30	30	30	30	60	60	60	60	60	60	60	60	60	60	60	60	
21	21	21	19	19	19	19	18	16	16	16	16	16	16	16	16	16	15	15	
550	550	550	564	564	564	564	564	564	606	606	606	606	606	606	606	606	606	506	
861	861	861	879	879	879	879	879	879	988	988	988	988	988	988	988	988	988	575	3750
30	30	30	30	30	30	30	30	30	60	60	60	60	60	60	60	60	60	without	
19	19	19	18	18	18	18	18	18	15	15	15	15	15	15	15	15	15	17	
524	524	524	524	564	564	564	564	564	564	586	586	586	586	586	586	586	586	586	
848	848	848	848	879	879	879	879	879	879	978	978	978	978	978	978	978	978	978	3500
without	without	without	without	30	30	30	30	30	30	60	60	60	60	60	60	60	60	60	
19	19	19	19	17	17	17	17	17	17	15	15	15	15	15	15	15	15	15	
502	518	518	518	557	557	564	564	564	564	576	576	576	576	576	576	576	576	576	
824	845	845	845	865	865	879	879	879	879	888	888	973	973	973	973	973	973	973	3250
without	without	without	without	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	
19	18	18	18	16	16	16	16	16	16	14	14	14	14	14	14	14	14	14	
500	514	514	514	553	553	553	553	562	562	562	576	576	576	576	576	576	576	576	
823	843	843	843	863	863	863	863	877	877	877	888	888	888	973	973	973	973	973	3000
without	without	without	without	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	
17	16	16	16	14	14	14	14	14	14	14	13	13	13	13	13	13	13	13	
500	512	512	512	512	519	519	519	519	519	519	574	574	574	574	574	574	574	574	
823	829	829	842	842	846	846	846	846	846	846	887	887	887	887	887	887	972	972	2750
without	without	without	without	without	without	without	without	without	without	without	30	30	30	30	30	30	30	30	
16	15	15	15	15	14	14	14	14	14	14	12	12	12	12	12	12	12	12	
500	500	488	488	488	519	519	519	519	519	519	519	567	567	567	567	567	567	567	
823	823	817	817	830	846	846	846	846	846	846	846	882	882	882	882	882	882	882	2500
without	without	without	without	without	without	without	without	without	without	without	without	30	30	30	30	30	30	30	
14	14	14	14	14	13	13	13	13	13	13	13	11	11	11	11	11	11	11	
460	460	460	476	476	476	519	519	519	519	519	519	519	553	553	553	553	553	553	
728	728	803	811	811	811	846	846	846	846	846	846	846	868	868	868	868	868	868	2250
without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	
14	14	14	13	13	13	11	11	11	11	11	11	11	10	10	10	10	10	10	
458	458	458	476	476	476	476	519	519	519	519	519	519	553	553	553	553	553	553	
727	727	727	811	811	811	811	846	846	846	846	846	846	868	868	868	868	868	868	2000
without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	
13	13	13	12	12	12	12	10	10	10	10	10	10	9	9	9	9	9	9	
6500		7000		7500		8000		8500		9000		9500		10000		10500		11000	

TAR 6.100 TKR 5.40 TKR 6.60 TAR 5.75 TAR 6.100 TAR 3.50

> 6000 wide with wind hooks

> 8000 wide and > 4000 high values with pressure shaft

Shutters for indoors, resistance to wind load class 2 in accordance with DIN EN 13241-1

Shutters for indoors, resistance to wind load class < 2 in accordance with DIN EN 13241-1

27. Table of reference values - 4020 Aluminium

S = Min. headroom
ET = Installation depth with emergency crank handle
UFU = Underfill height
T = Opening time of the shutter system approx. in seconds

Height		Shaft diameter 159 mm															177 mm		
	S	740	740	740	740	740	740	740	740	740	740	740	740	740	740	740	752	752	752
	ET	853	853	853	853	853	853	928	928	941	941	941	941	941	941	941	947	961	961
8000	UFU	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80
	T	34	34	34	34	34	34	34	34	34	34	34	34	34	34	34	33	33	33
	S	740	740	740	740	740	740	740	740	740	740	740	740	740	740	740	752	752	752
	ET	852	853	853	853	853	853	928	928	941	941	941	941	941	941	941	947	947	961
7750	UFU	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80
	T	33	33	33	33	33	33	33	33	33	33	33	33	33	33	33	32	32	32
	S	740	740	740	740	740	740	740	740	740	740	740	740	740	740	740	742	742	742
	ET	852	853	853	853	853	853	928	928	928	941	941	941	941	941	941	942	942	956
7500	UFU	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80
	T	32	32	32	32	32	32	32	32	32	32	32	32	32	32	32	31	31	31
	S	720	720	720	720	720	720	720	720	720	720	720	720	720	720	720	720	704	704
	ET	842	843	843	843	843	843	843	918	918	931	931	931	931	931	931	931	923	923
7250	UFU	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	60	60
	T	32	32	32	32	32	32	32	32	32	32	32	32	32	32	32	32	31	31
	S	700	700	700	700	700	700	700	700	700	700	700	700	700	700	700	700	704	704
	ET	832	832	833	833	833	833	833	908	908	908	921	921	921	921	921	921	923	923
7000	UFU	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60
	T	31	31	31	31	31	31	31	31	31	31	31	31	31	31	31	31	30	30
	S	700	700	700	700	700	700	700	700	700	700	700	700	700	700	700	700	704	704
	ET	832	832	833	833	833	833	833	833	908	908	921	921	921	921	921	921	923	923
6750	UFU	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60
	T	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	29	29
	S	700	700	700	700	700	700	700	700	700	700	700	700	700	700	700	700	702	702
	ET	832	832	833	833	833	833	833	833	908	908	908	921	921	921	921	921	922	922
6500	UFU	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60
	T	29	29	29	29	29	29	29	29	29	29	29	29	29	29	29	29	28	28
	S	700	700	700	700	700	700	700	700	700	700	700	700	700	700	700	700	702	702
	ET	832	832	833	833	833	833	833	833	908	908	908	908	921	921	921	921	922	922
6250	UFU	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60
	T	28	28	28	28	28	28	28	28	28	28	28	28	28	28	28	28	27	27
	S	700	700	700	700	700	700	700	700	700	700	700	700	700	700	700	700	700	696
	ET	832	832	832	833	833	833	833	833	908	908	908	908	908	921	921	921	921	919
6000	UFU	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60
	T	27	27	27	27	27	27	27	27	27	27	27	27	27	27	27	27	27	26
	S	670	670	670	670	670	670	670	670	670	670	670	670	670	670	670	670	670	664
	ET	817	817	817	818	818	818	818	818	893	893	893	893	893	893	906	906	906	903
5750	UFU	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60
	T	27	27	27	27	27	27	27	27	27	27	27	27	27	27	27	27	27	26
	S	614	614	614	614	614	614	614	614	614	614	614	614	614	614	614	614	614	626
	ET	804	804	804	804	805	805	805	805	880	880	880	880	880	880	880	893	893	899
5500	UFU	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60
	T	27	27	27	27	27	27	27	27	27	27	27	27	27	27	27	27	27	25
	S	614	614	614	614	614	614	614	614	614	614	614	614	614	614	614	614	614	626
	ET	804	804	804	804	805	805	805	805	805	880	880	880	880	880	880	880	893	899
5250	UFU	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60
	T	25	25	25	25	25	25	25	25	25	25	25	25	25	25	25	25	25	24
	S	614	614	614	614	614	614	614	614	614	614	614	614	614	614	614	614	614	614
	ET	804	804	804	804	804	805	805	805	805	880	880	880	880	880	880	880	880	893
5000	UFU	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60
	T	24	24	24	24	24	24	24	24	24	24	24	24	24	24	24	24	24	24
	S	614	614	614	614	614	614	614	614	614	614	614	614	614	614	614	614	614	614
	ET	804	804	804	804	804	804	805	805	805	805	880	880	880	880	880	880	880	880
4750	UFU	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60
	T	23	23	23	23	23	23	23	23	23	23	23	23	23	23	23	23	23	23
	S	610	610	610	610	610	610	610	610	610	610	610	610	610	610	610	610	610	610
	ET	802	802	802	802	802	802	803	803	803	803	803	878	878	878	878	878	878	878
4500	UFU	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60
	T	22	22	22	22	22	22	22	22	22	22	22	22	22	22	22	22	22	22
	S	574	574	574	574	574	574	574	574	574	574	574	574	574	574	574	574	574	574
	ET	784	784	784	784	784	784	784	785	785	785	785	785	860	860	860	860	860	860
4250	UFU	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30
	T	22	22	22	22	22	22	22	22	22	22	22	22	22	22	22	22	22	22
	S	566	566	566	566	566	566	566	566	566	566	566	566	566	566	566	566	566	566
	ET	780	780	780	780	780	780	780	780	781	781	781	781	781	781	856	856	856	856
4000	UFU	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30
	T	21	21	21	21	21	21	21	21	21	21	21	21	21	21	21	21	21	21
	S	566	566	566	566	566	566	566	566	566	566	566	566	566	566	566	566	566	566
	ET	780	780	780	780	780	780	780	780	781	781	781	781	781	781	781	781	856	856
3750	UFU	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30
	T	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20
	S	566	566	566	566	566	566	566	566	566	566	566	566	566	566	566	566	566	566
	ET	780	780	780	780	780	780	780	780	781	781	781	781	781	781	781	781	781	856
3500	UFU	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30
	T	18	18	18	18	18	18	18	18	18	18	18	18	18	18	18	18	18	18
	S	560	560	560	560	560	560	560	560	560	560	560	560	560	560	560	560	560	560
	ET	777	777	777	777	777	777	777	777	778	778	778	778	778	778	778	778	778	778
3250	UFU	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30
	T	17	17	17	17	17	17	17	17	17	17	17	17	17	17	17	17	17	17
	S	520	520	520	520	520	520	520	520	520	520	520	520	520	520	520	520	520	520
	ET	757	757	757	757	757	757	757	757	757	758	758	758	758	758	758	758	758	758
3000	UFU	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without
	T	17	17	17	17	17	17	17	17	17	17	17	17	17	17	17	17	17	17
	S	514	514	514	514	514	514	514	514	514	514	514	514	514	514	514	514	514	514
	ET	754	754	754	754	754	754	754	754	754	754	755	755	755	755	755	755	755	755
2750	UFU	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without
	T	16	16	16	16	16	16	16	16	16	16	16	16	16	16	16	16	16	16
	S	514	514	514	514	514	514	514	514	514	514	514	514	514	514	514	514	514	514
	ET	754	754	754	754	754	754	754	754	754	754	754	754	755	755	755	755	755	755
2500	UFU	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without
	T	14	14	14	14	14	14	14	14	14	14	14	14	14	14	14	14	14	14
	S	514	514	514	514	514	514	514	514	514	514	514	514	514	514	514	514	514	514
	ET	754	754	754	754	754	754	754	754	754	754	754	754	754	754	754	755	755	755
2250	UFU	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without
	T	13	13	13	13	13	13	13	13	13	13	13	13	13	13	13	13	13	13
	S	504	504	504	504	504	504	504	504	504	504	504	504	504	504	504	504	504	498
	ET	746	746	746	746	746	746	746	746	746	746	746	746	746	746	746	746	746	747
2000	UFU	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without
	T	12	12	12	12	12	12	12	12	12	12	12	12	12	12	12	12	12	12
Width >		2000		2500		3000		3500		4000		4500		5000		5500		6000	

Shutters for indoors, resistance to wind load class 2 in accordance with DIN EN 13241-1

TAR 3.42
TAR 3.30
TAR 1.14
TAR 2.22

27. Table of reference values - 4020 Aluminium

Advice: The tables of standard values are only for standard shutters.

219 mm						273 mm					298 mm				323 mm				Height
744	744	744	744	744	744	990	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	
957	957	957	1042	1042	1042	690	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	
80	80	80	80	80	80	80	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	8000
31	31	31	31	31	31	28	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	
752	744	744	744	744	744	990	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	
961	957	957	957	1042	1042	690	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	7750
80	80	80	80	80	80	80	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	
32	30	30	30	30	30	27	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	
742	744	744	744	744	744	990	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	
956	957	957	957	957	1042	690	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	7500
80	80	80	80	80	80	80	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	
31	29	29	29	29	29	26	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	
704	744	744	744	744	744	966	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	
937	957	957	957	957	957	1048	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	7250
60	80	80	80	80	80	80	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	
31	28	28	28	28	28	26	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	
704	740	740	740	740	740	946	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	
923	955	955	955	955	955	1038	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	7000
60	80	80	80	80	80	80	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	
30	27	27	27	27	27	26	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	
704	740	740	740	740	740	950	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	
923	955	955	955	955	955	955	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	6750
60	80	80	80	80	80	80	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	
29	26	26	26	26	26	26	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	
702	702	720	720	720	720	950	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	
922	922	945	945	945	945	955	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	6500
60	60	80	80	80	80	80	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	
28	28	26	26	26	26	25	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	
702	702	700	700	700	700	910	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	
922	922	935	935	935	935	935	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	6250
60	60	60	60	60	60	60	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	
27	27	26	26	26	26	26	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	
696	696	700	700	700	700	910	946	946	946	946	946	946	946	946	946	990	990	990	
919	919	935	935	935	935	935	1038	1038	1038	1038	1038	646	646	646	646	690	690	690	6000
60	60	60	60	60	60	60	80	80	80	80	80	80	80	80	80	80	80	80	
26	26	25	25	25	25	25	22	22	22	22	22	22	22	22	22	20	20	20	
664	664	700	700	700	700	910	910	946	946	946	946	946	946	946	946	962	962	962	
903	903	935	935	935	935	935	935	1038	1038	1038	1038	1038	646	646	646	662	662	662	5750
60	60	60	60	60	60	60	60	80	80	80	80	80	80	80	80	80	80	80	
26	26	24	24	24	24	24	24	21	21	21	21	21	21	21	21	20	20	20	
626	626	626	662	662	662	880	880	916	916	916	916	916	916	916	916	916	906	906	
899	899	899	931	931	931	935	935	1038	1038	1038	1038	1038	1038	1038	646	646	636	636	5500
60	60	60	60	60	60	60	60	80	80	80	80	80	80	80	80	80	80	80	
25	25	25	23	23	23	23	23	20	20	20	20	20	20	20	20	20	20	20	
626	626	626	662	662	662	872	872	870	870	870	870	870	870	870	870	870	906	906	
899	899	899	931	931	931	931	931	1015	1015	1015	1015	1015	1015	1015	1015	1015	636	636	5250
60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	80	80	
24	24	24	22	22	22	22	22	20	20	20	20	20	20	20	20	20	19	19	
626	626	626	658	658	658	872	872	872	870	870	870	870	870	870	870	870	906	906	
899	899	899	929	929	929	931	931	931	1015	1015	1015	1015	1015	1015	1015	1015	636	636	5000
60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	80	80	
23	23	23	21	21	21	21	21	21	19	19	19	19	19	19	19	19	18	18	
626	626	626	626	612	612	842	842	842	870	870	870	870	870	870	870	870	870	906	
899	899	899	899	892	906	916	916	916	1015	1015	1015	1015	1015	1015	1015	1015	1015	636	4750
60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	80	
22	22	22	22	21	21	20	20	20	18	18	18	18	18	18	18	18	18	17	
620	620	620	620	612	612	822	822	822	870	870	870	870	870	870	870	870	870	906	
896	896	896	896	892	892	906	906	906	930	1015	1015	1015	1015	1015	1015	1015	1015	636	4500
60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	80	
21	21	21	21	20	20	20	20	20	17	17	17	17	17	17	17	17	17	16	
574	578	578	578	612	612	822	822	822	822	870	870	870	870	870	870	870	870	870	
860	875	875	875	892	892	892	906	906	906	930	930	1015	1015	1015	1015	1015	1015	1015	4250
30	30	30	30	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	
22	21	21	21	19	19	19	19	19	19	16	16	16	16	16	16	16	16	16	
566	578	578	578	578	612	822	822	822	822	870	870	870	870	870	870	870	870	870	
856	875	875	875	875	892	892	906	906	906	930	930	930	1015	1015	1015	1015	1015	1015	4000
30	30	30	30	30	60	60	60	60	60	60	60	60	60	60	60	60	60	60	
21	20	20	20	20	18	18	18	18	18	15	15	15	15	15	15	15	15	15	
566	572	572	572	572	610	610	610	610	610	610	610	610	610	610	610	610	610	610	
856	859	859	872	872	891	891	891	891	891	891	905	905	905	905	905	905	990	990	3750
30	30	30	30	30	60	60	60	60	60	60	60	60	60	60	60	60	60	60	
20	19	19	19	19	17	17	17	17	17	17	15	15	15	15	15	15	15	15	
566	566	572	572	572	572	562	562	562	562	562	562	610	610	610	610	610	610	610	
856	856	859	859	872	872	867	867	867	867	867	867	905	905	905	905	905	905	905	3500
30	30	30	30	30	30	30	30	30	30	30	30	60	60	60	60	60	60	60	
18	18	17	17	17	17	17	17	17	17	17	17	14	14	14	14	14	14	14	
560	560	572	572	572	572	562	562	562	562	562	562	606	606	606	606	606	606	606	
853	853	859	859	859	859	867	867	867	867	867	867	903	903	903	903	903	903	903	3250
30	30	30	30	30	30	30	30	30	30	30	30	60	60	60	60	60	60	60	
17	17	16	16	16	16	15	15	15	15	15	15	13	13	13	13	13	13	13	
520	520	520	524	524	524	560	560	560	560	560	560	560	606	606	606	606	606	606	
758	758	758	835	835	835	866	866	866	866	866	866	866	903	903	903	903	903	903	3000
without	without	without	without	without	without	30	30	30	30	30	30	30	60	60	60	60	60	60	
17	17	17	16	16	16	14	14	14	14	14	14	14	12	12	12	12	12	12	
514	514	514	524	524	524	524	560	560	560	560	560	560	606	606	606	606	606	606	
755	755	755	835	835	835	835	866	866	866	866	866	866	903	903	903	903	903	903	2750
without	without	without	without	without	without	without	30	30	30	30	30	30	60	60	60	60	60	60	
16	16	16	15	15	15	15	13	13	13	13	13	13	11	11	11	11	11	11	
514	514	514	514	518	518	518	560	560	560	560	560	560	560	565	565	565	565	565	
755	755	755	755	832	832	832	866	866	866	866	866	866	866	880	880	880	880	880	2500
without	without	without	without	without	without	without	30	30	30	30	30	30	30	30	30	30	30	30	
14	14	14	14	14	14	14	12	12	12	12	12	12	12	11	11	11	11	11	
514	514	514	514	524	524	524	524	544	544	544	544	544	544	560	560	565	565	565	
755	755	755	755	760	760	835	835	858	858	858	858	858	858	866	866	880	880	880	2250
without	without	without	without	without	without	without	without	without	without	without	without	without	without	30	30	30	30	30	
13	13	13	13	12	12	12	12	11	11	11	11	11	11	10	10	10	10	10	
498	498	498	498	498	494	494	494	510	510	510	510	510	510	510	560	560	560	560	
747	747	747	747	747	745	745	745	828	828	841	841	841	841	841	866	866	866	866	2000
without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	30	30	30	30	
12	12	12	12	12	11	11	11	10	10	10	10	10	10	9	9	9	9		
6500		7000		7500		8000		8500		9000	9500		10000		10500		11000		

TAR 5.75 TAR 6.100 TKR 5.40 TAR 2.50 TAR 3.50

> 6000 wide with wind hooks

> 8000 wide and > 4000 high values with pressure shaft

Shutters for indoors, resistance to wind load class 2 in accordance with DIN EN 13241-1

Shutters for indoors, resistance to wind load class < 2 in accordance with DIN EN 13241-1

28. Table of reference values - 6010 Aluminium

S = Min. headroom
ET = Installation depth with emergency crank handle
UFU = Underfill height
T = Opening time of the shutter system approx. in seconds

Shaft diameter 159 mm

Height		Width > 2000			2500			3000			3500			4000			4500			5000			5500			6000			Height
8000	S	694	694	694	694	694	694	694	694	694	694	694	694	694	694	694	694	694	694	694	694	694	694	694	694	694	694	694	8000
	ET	829	829	829	829	829	830	830	830	830	830	830	830	830	830	830	830	830	905	905	905	905	905	905	905	905	905	905	
	UFU	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	
	T	36	36	36	36	36	36	36	36	36	36	36	36	36	36	36	36	36	36	36	36	36	36	36	36	36	36	36	
7750	S	668	668	668	668	668	668	668	668	668	668	668	668	668	668	668	668	668	668	668	668	668	668	668	668	668	668	668	7750
	ET	816	816	816	816	816	816	817	817	817	817	817	817	817	817	817	817	817	892	892	892	892	892	892	892	892	892	892	
	UFU	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	
	T	36	36	36	36	36	36	36	36	36	36	36	36	36	36	36	36	36	36	36	36	36	36	36	36	36	36	36	
7500	S	668	668	668	668	668	668	668	668	668	668	668	668	668	668	668	668	668	668	668	668	668	668	668	668	668	668	668	7500
	ET	816	816	816	816	816	816	817	817	817	817	817	817	817	817	817	817	817	892	892	892	892	892	892	892	892	892	892	
	UFU	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	
	T	35	35	35	35	35	35	35	35	35	35	35	35	35	35	35	35	35	35	35	35	35	35	35	35	35	35	35	
7250	S	668	668	668	668	668	668	668	668	668	668	668	668	668	668	668	668	668	668	668	668	668	668	668	668	668	668	668	7250
	ET	816	816	816	816	816	816	816	817	817	817	817	817	817	817	817	817	817	892	892	892	892	892	892	892	892	892	892	
	UFU	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	
	T	34	34	34	34	34	34	34	34	34	34	34	34	34	34	34	34	34	34	34	34	34	34	34	34	34	34	34	
7000	S	668	668	668	668	668	668	668	668	668	668	668	668	668	668	668	668	668	668	668	668	668	668	668	668	668	668	668	7000
	ET	816	816	816	816	816	816	816	817	817	817	817	817	817	817	817	817	817	892	892	892	892	892	892	892	892	892	892	
	UFU	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	
	T	33	33	33	33	33	33	33	33	33	33	33	33	33	33	33	33	33	33	33	33	33	33	33	33	33	33	33	
6750	S	668	668	668	668	668	668	668	668	668	668	668	668	668	668	668	668	668	668	668	668	668	668	668	668	668	668	668	6750
	ET	816	816	816	816	816	816	816	816	817	817	817	817	817	817	817	817	817	817	817	892	892	892	892	892	892	892	892	
	UFU	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	
	T	32	32	32	32	32	32	32	32	32	32	32	32	32	32	32	32	32	32	32	32	32	32	32	32	32	32	32	
6500	S	638	638	638	638	638	638	638	638	638	638	638	638	638	638	638	638	638	638	638	638	638	638	638	638	638	638	638	6500
	ET	801	801	801	801	801	801	801	801	802	802	802	802	802	802	802	802	802	802	802	802	877	877	877	877	877	877	877	
	UFU	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	
	T	32	32	32	32	32	32	32	32	32	32	32	32	32	32	32	32	32	32	32	32	32	32	32	32	32	32	32	
6250	S	638	638	638	638	638	638	638	638	638	638	638	638	638	638	638	638	638	638	638	638	638	638	638	638	638	638	638	6250
	ET	801	801	801	801	801	801	801	801	801	802	802	802	802	802	802	802	802	802	802	802	802	877	877	877	877	877	877	
	UFU	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	
	T	31	31	31	31	31	31	31	31	31	31	31	31	31	31	31	31	31	31	31	31	31	31	31	31	31	31	31	
6000	S	638	638	638	638	638	638	638	638	638	638	638	638	638	638	638	638	638	638	638	638	638	638	638	638	638	638	638	6000
	ET	801	801	801	801	801	801	801	801	801	801	802	802	802	802	802	802	802	802	802	802	802	802	877	877	877	877	877	
	UFU	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	
	T	29	29	29	29	29	29	29	29	29	29	29	29	29	29	29	29	29	29	29	29	29	29	29	29	29	29	29	
5750	S	638	638	638	638	638	638	638	638	638	638	638	638	638	638	638	638	638	638	638	638	638	638	638	638	638	638	638	5750
	ET	801	801	801	801	801	801	801	801	801	801	801	802	802	802	802	802	802	802	802	802	802	802	802	802	802	802	802	
	UFU	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	
	T	28	28	28	28	28	28	28	28	28	28	28	28	28	28	28	28	28	28	28	28	28	28	28	28	28	28	28	
5500	S	608	608	608	608	608	608	608	608	608	608	608	608	608	608	608	608	608	608	608	608	608	608	608	608	608	608	608	5500
	ET	801	801	801	801	801	801	801	801	801	801	801	801	802	802	802	802	802	802	802	802	802	802	802	802	802	802	802	
	UFU	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	
	T	27	27	27	27	27	27	27	27	27	27	27	27	27	27	27	27	27	27	27	27	27	27	27	27	27	27	27	
5250	S	570	570	570	570	570	570	570	570	570	570	570	570	570	570	570	570	570	570	570	570	570	570	570	570	570	570	570	5250
	ET	782	782	782	782	782	782	782	782	782	782	782	782	782	782	782	782	782	783	783	783	783	783	783	783	783	783	783	
	UFU	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	
	T	27	27	27	27	27	27	27	27	27	27	27	27	27	27	27	27	27	27	27	27	27	27	27	27	27	27	27	
5000	S	570	570	570	570	570	570	570	570	570	570	570	570	570	570	570	570	570	570	570	570	570	570	570	570	570	570	570	5000
	ET	782	782	782	782	782	782	782	782	782	782	782	782	782	782	782	782	782	783	783	783	783	783	783	783	783	783	783	
	UFU	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	
	T	26	26	26	26	26	26	26	26	26	26	26	26	26	26	26	26	26	26	26	26	26	26	26	26	26	26	26	
4750	S	570	570	570	570	570	570	570	570	570	570	570	570	570	570	570	570	570	570	570	570	570	570	570	570	570	570	570	4750
	ET	782	782	782	782	782	782	782	782	782	782	782	782	782	782	782	782	782	782	783	783	783	783	783	783	783	783	783	
	UFU	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	
	T	25	25	25	25	25	25	25	25	25	25	25	25	25	25	25	25	25	25	25	25	25	25	25	25	25	25	25	
4500	S	570	570	570	570	570	570	570	570	570	570	570	570	570	570	570	570	570	570	570	570	570	570	570	570	570	570	570	4500
	ET	782	782	782	782	782	782	782	782	782	782	782	782	782	782	782	782	782	782	783	783	783	783	783	783	783	783	783	
	UFU	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	
	T	23	23	23	23	23	23	23	23	23	23	23	23	23	23	23	23	23	23	23	23	23	23	23	23	23	23	23	
4250	S	524	524	524	524	524	524	524	524	524	524	524	524	524	524	524	524	524	524	524	524	524	524	524	524	524	524	524	4250
	ET	759	759	759	759	759	759	759	759	759	759	759	759	759	759	759	759	759	759	760	760	760	760	760	760	760	760	760	
	UFU	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	
	T	24	24	24	24	24	24	24	24	24	24	24	24	24	24	24	24	24	24	24	24	24	24	24	24	24	24	24	
4000	S	524	524	524	524	524	524	524	524	524	524	524	524	524	524	524	524	524	524	524	524	524	524	524	524	524	524	524	4000
	ET	759	759	759	759	759	759	759	759	759	759	759	759	759	759	759	759	759	759	759	760	760	760	760	760	760	760	760	
	UFU	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	
	T	22	22	22	22	22	22	22	22	22	22	22	22	22	22	22	22	22	22	22	22	22	22	22	22	22	22	22	
3750	S	524	524	524	524	524	524	524	524	524	524	524	524	524	524	524	524	524	524	524	524	524	524	524	524	524	524	524	3750
	ET	759	759	759	759	759	759	759	759	759	759	759	759	759	759	759	759	759	759	759	759	759	760	760	760	760	760	760	
	UFU	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	
	T	21	21	21	21	21	21	21	21	21	21	21	21	21	21	21	21	21	21	21	21	21	21	21	21	21	21	21	
3500	S	524	524	524	524	524	524	524	524	524	524	524	524	524	524	524	524	524	524	524	524	524	524	524	524	524	524	524	3500
	ET	759	759	759	759	759	759	759	759	759	759	759	759	759	759	759	759	759	759	759	759	759	759	759	759	759	759	759	
	UFU	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	
	T	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	
3250	S	498	498	498	498	498	498	498	498	498	498	498	498	498	498	498	498	498	498	498	498	498	498	498	498	498	498	498	3250
	ET	740	740	740	740	740	740	740	740	740	740	740	740	740	740	740	740	740	740	740	740	740	740	740	740	740	740	740	
	UFU	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	
	T	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	
3000	S	498	498	498	498	498	498	498	498	498	498	498	498	498	498	498	498	498	498	498	498	498	498	498	498	498	498	498	3000
	ET	740	740	740	740	740	740	740	740	740	740	740	740	740	740	740	740	740	740	740	740	740	740	740	740	740	740	740	
	UFU	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	
	T	18	18	18	18	18	18	18	18	18	18	18	18	18	18	18	18	18	18	18	18	18	18	18	18	18	18	18	
2750	S	498	498	498	498	498	498	498	498	498	498	498	498	498	498	498	498	498	498	498	498	498	498	498	498	498	498	498	2750
	ET	740	740	740	740	740	740	740	740	740	740	740	740	740	740	740	740	740	740	740	740	740	740	740	740	740	740	740	
	UFU	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	
	T	16	16	16	16	16	16	16	16	16	16	16	16	16	16	16	16	16	16	16	16	16	16	16	16	16	16	16	
2500	S	498	498	498	498	498	498	498	498	498	498	498	498	498	498	498	498	498	498	498	498	498	498	498	498	498	498	498	2500
	ET	740	740	740	740	740	740	740	740	740	740	740	740	740	740	740	740	740	740	740	740	740	740	740	740	740	740	740	
	UFU	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	
	T	15	15	15	15	15	15	15	15	15	15	15	15	15	15	15	15	15	15	15	15	15	15	15	15	15	15	15	
2250	S	478	478	478	478	478	478	478	478	478	478	478	478	478	478	478	478	478	478	478	478	478	478	478	478	478	478	478	2250
	ET	720	720	720	720	720	720	720	720	720	720	720	720	720	720	720	720	720	720	720	720	720	720	720	720	720	720	720	
	UFU	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	
	T	15	15	15	15	15	15	15	15	15	15	15	15	15	15	15	15	15	15	15	15	15	15	15	15	15	15	15	
2000	S	478	478	478	478	478	478	478	478	478	478	478	478	478	478	478	478	478	478	478	478	478	478	478	478	478	478	478	2000
	ET	720	720	720	720	720	720	720	720	720	720	720	720	720	720	720	720	720	720	720	720	720	720	720	720	720	720	720	
	UFU	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	
	T	13	13	13	13	13	13	13	13	13	13	13	13	13	13	13	13	13	13	13	13	13	13	13	13	13	13	13	

> 5500 wide with wind hooks

Shutters for indoors, resistance to wind load class 2 in accordance with DIN EN 13241-1

TAR 3.30
TAR 2.22
TAR 1.14

29. Table of reference values - 6010 Steel

Advice: The tables of standard values are only for standard shutters.

Height		Shaft diameter 159 mm												177 mm			219 mm		Height
	S	694	694	694	694	694	694	694	694	694	694	694	694	686	686	686	688	688	
	ET	905	905	918	918	918	918	918	918	918	918	932	932	928	928	928	1014	1014	
8000	UFU	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	8000
	T	36	36	36	36	36	36	36	36	36	36	36	36	36	36	36	34	34	
	S	668	668	668	668	668	668	668	668	668	668	668	668	686	686	686	688	688	
	ET	892	892	892	905	905	905	905	905	905	905	919	919	928	928	928	1014	1014	
7750	UFU	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	7750
	T	36	36	36	36	36	36	36	36	36	36	36	36	35	35	35	32	32	
	S	668	668	668	668	668	668	668	668	668	668	668	668	668	676	676	676	688	
	ET	892	892	892	905	905	905	905	905	905	905	905	919	919	923	923	923	1014	
7500	UFU	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	7500
	T	35	35	35	35	35	35	35	35	35	35	35	35	35	34	34	34	31	
	S	668	668	668	668	668	668	668	668	668	668	668	668	668	676	676	676	688	
	ET	892	892	892	905	905	905	905	905	905	905	905	919	919	923	923	923	1014	
7250	UFU	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	7250
	T	34	34	34	34	34	34	34	34	34	34	34	34	34	33	33	33	30	
	S	668	668	668	668	668	668	668	668	668	668	668	668	668	654	654	654	688	
	ET	817	892	892	892	905	905	905	905	905	905	905	905	919	912	912	912	1014	
7000	UFU	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	7000
	T	33	33	33	33	33	33	33	33	33	33	33	33	33	33	33	33	29	
	S	668	668	668	668	668	668	668	668	668	668	668	668	668	654	654	654	654	
	ET	817	892	892	892	905	905	905	905	905	905	905	905	905	912	912	912	912	
6750	UFU	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	6750
	T	32	32	32	32	32	32	32	32	32	32	32	32	32	31	31	31	31	
	S	638	638	638	638	638	638	638	638	638	638	638	638	638	638	654	654	654	
	ET	802	877	877	877	877	890	890	890	890	890	890	890	890	890	912	912	912	
6500	UFU	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	6500
	T	32	32	32	32	32	32	32	32	32	32	32	32	32	32	30	30	30	
	S	638	638	638	638	638	638	638	638	638	638	638	638	638	638	654	654	654	
	ET	802	802	877	877	877	890	890	890	890	890	890	890	890	890	912	912	912	
6250	UFU	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	6250
	T	31	31	31	31	31	31	31	31	31	31	31	31	31	31	29	29	29	
	S	638	638	638	638	638	638	638	638	638	638	638	638	638	638	654	654	654	
	ET	802	802	877	877	877	877	890	890	890	890	890	890	890	890	912	912	912	
6000	UFU	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	6000
	T	29	29	29	29	29	29	29	29	29	29	29	29	29	29	28	28	28	
	S	638	638	638	638	638	638	638	638	638	638	638	638	638	638	626	626	626	
	ET	802	802	877	877	877	877	890	890	890	890	890	890	890	890	898	898	898	
5750	UFU	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	5750
	T	28	28	28	28	28	28	28	28	28	28	28	28	28	28	28	28	28	
	S	608	608	608	608	608	608	608	608	608	608	608	608	608	608	608	596	596	
	ET	802	802	802	877	877	877	877	890	890	890	890	890	890	890	890	898	898	
5500	UFU	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	5500
	T	27	27	27	27	27	27	27	27	27	27	27	27	27	27	27	27	27	
	S	570	570	570	570	570	570	570	570	570	570	570	570	570	570	570	596	596	
	ET	783	783	783	858	858	858	858	858	858	871	871	871	871	871	871	898	898	
5250	UFU	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	60	60	5250
	T	27	27	27	27	27	27	27	27	27	27	27	27	27	27	27	25	25	
	S	570	570	570	570	570	570	570	570	570	570	570	570	570	570	570	596	596	
	ET	783	783	783	783	858	858	858	858	858	871	871	871	871	871	871	884	898	
5000	UFU	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	60	60	5000
	T	26	26	26	26	26	26	26	26	26	26	26	26	26	26	26	24	24	
	S	570	570	570	570	570	570	570	570	570	570	570	570	570	570	570	572	572	
	ET	783	783	783	783	858	858	858	858	858	871	871	871	871	871	871	872	872	
4750	UFU	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	4750
	T	25	25	25	25	25	25	25	25	25	25	25	25	25	25	25	24	24	
	S	570	570	570	570	570	570	570	570	570	570	570	570	570	570	570	570	572	
	ET	783	783	783	783	783	858	858	858	858	858	871	871	871	871	871	871	872	
4500	UFU	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	4500
	T	23	23	23	23	23	23	23	23	23	23	23	23	23	23	23	23	23	
	S	524	524	524	524	524	524	524	524	524	524	524	524	524	524	524	524	572	
	ET	760	760	760	760	760	760	835	835	835	835	835	848	848	848	848	848	872	
4250	UFU	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	30	4250
	T	24	24	24	24	24	24	24	24	24	24	24	24	24	24	24	24	21	
	S	524	524	524	524	524	524	524	524	524	524	524	524	524	524	524	524	572	
	ET	759	760	760	760	760	760	835	835	835	835	835	835	848	848	848	848	872	
4000	UFU	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	30	4000
	T	22	22	22	22	22	22	22	22	22	22	22	22	22	22	22	22	20	
	S	524	524	524	524	524	524	524	524	524	524	524	524	524	524	524	524	524	
	ET	759	760	760	760	760	760	760	835	835	835	835	835	835	848	848	848	848	
3750	UFU	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	3750
	T	21	21	21	21	21	21	21	21	21	21	21	21	21	21	21	21	21	
	S	524	524	524	524	524	524	524	524	524	524	524	524	524	524	524	524	524	
	ET	759	759	760	760	760	760	760	760	835	835	835	835	835	835	835	848	848	
3500	UFU	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	3500
	T	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	
	S	498	498	498	486	486	486	486	486	486	486	486	486	486	486	486	486	486	
	ET	740	740	740	741	741	741	741	741	741	741	816	816	816	816	816	816	829	
3250	UFU	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	3250
	T	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	
	S	498	498	498	498	486	486	486	486	486	486	486	486	486	486	486	486	486	
	ET	740	740	740	740	741	741	741	741	741	741	741	816	816	816	816	816	816	
3000	UFU	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	3000
	T	18	18	18	18	18	18	18	18	18	18	18	18	18	18	18	18	18	
	S	498	498	498	498	486	486	486	486	486	486	486	486	486	486	486	486	486	
	ET	740	740	740	740	741	741	741	741	741	741	741	741	816	816	816	816	816	
2750	UFU	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	2750
	T	16	16	16	16	16	16	16	16	16	16	16	16	16	16	16	16	16	
	S	498	498	498	498	498	486	486	486	486	486	486	486	486	486	486	486	486	
	ET	740	740	740	740	740	741	741	741	741	741	741	741	741	741	816	816	816	
2500	UFU	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	2500
	T	15	15	15	15	15	15	15	15	15	15	15	15	15	15	15	15	15	
	S	478	478	478	478	478	478	478	446	446	446	446	446	446	446	446	446	446	
	ET	720	720	720	720	720	720	720	721	721	721	721	721	721	721	721	721	796	
2250	UFU	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	2250
	T	15	15	15	15	15	15	15	15	15	15	15	15	15	15	15	15	15	
	S	478	478	478	478	478	478	478	478	446	446	446	446	446	446	446	446	446	
	ET	720	720	720	720	720	720	720	720	721	721	721	721	721	721	721	721	721	
2000	UFU	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	2000
	T	13	13	13	13	13	13	13	13	13	13	13	13	13	13	13	13	13	
Width >		**2000**		**2500**		**3000**		**3500**		**4000**		**4500**		**5000**		**5500**		**6000**	

■ > 5500 wide with wind hooks

Shutters for indoors, resistance to wind load class 2 in accordance with DIN EN 13241-1

TAR 5.75
TAR 3.50
TAR 3.42
TAR 3.30
TAR 2.22
TAR 1.14

30. Table of reference values - rolling grating steel, TG-S and TG-E

S = Min. headroom
ET = Installation depth with emergency crank handle
UFU = Underfill height
T = Opening time of the shutter system approx. in seconds

Height		Shaft diameter 159 mm											177 mm			219 mm			
	S	642	642	642	642	642	642	642	642	642	642	642	700	700	700	716	716	716	716
	ET	879	879	892	892	892	892	892	892	892	906	906	935	1020	1020	626	626	626	626
8000	UFU	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80
	T	39	39	39	39	39	39	39	39	39	39	39	35	35	35	32	32	32	32
	S	620	620	620	620	620	620	620	620	620	620	620	670	674	674	716	716	716	716
	ET	868	868	881	881	881	881	881	881	881	895	895	920	1007	1007	626	626	626	626
7750	UFU	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80
	T	39	39	39	39	39	39	39	39	39	39	39	36	35	35	31	31	31	31
	S	620	620	620	620	620	620	620	620	620	620	620	670	674	674	674	700	700	700
	ET	868	868	868	881	881	881	881	881	881	881	895	920	922	1007	1007	610	610	610
7500	UFU	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80
	T	38	38	38	38	38	38	38	38	38	38	38	35	34	34	34	31	31	31
	S	620	620	620	620	620	620	620	620	620	620	620	654	674	674	674	700	700	700
	ET	868	868	868	881	881	881	881	881	881	881	895	912	922	922	1007	610	610	610
7250	UFU	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80
	T	37	37	37	37	37	37	37	37	37	37	37	35	33	33	33	30	30	30
	S	620	620	620	620	620	620	620	620	620	620	620	654	668	668	668	676	676	676
	ET	868	868	868	881	881	881	881	881	881	881	881	912	919	919	1004	1008	586	586
7000	UFU	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80
	T	35	35	35	35	35	35	35	35	35	35	35	33	32	32	32	30	30	30
	S	610	610	610	610	610	610	610	610	610	610	610	654	668	668	668	676	676	676
	ET	788	863	863	863	876	876	876	876	876	876	876	912	919	919	919	1008	1008	586
6750	UFU	60	60	60	60	60	60	60	60	60	60	60	80	80	80	80	80	80	80
	T	35	35	35	35	35	35	35	35	35	35	35	32	31	31	31	29	29	29
	S	592	592	592	592	592	592	592	592	592	592	592	648	648	648	648	648	676	676
	ET	779	854	854	854	867	867	867	867	867	867	867	909	909	909	909	909	1008	586
6500	UFU	60	60	60	60	60	60	60	60	60	60	60	80	80	80	80	80	80	80
	T	34	34	34	34	34	34	34	34	34	34	34	31	31	31	31	31	28	28
	S	592	592	592	592	592	592	592	592	592	592	592	636	636	634	634	634	676	676
	ET	779	854	854	854	854	867	867	867	867	867	867	903	903	902	902	902	1008	1008
6250	UFU	60	60	60	60	60	60	60	60	60	60	60	80	80	80	80	80	80	80
	T	33	33	33	33	33	33	33	33	33	33	33	31	31	30	30	30	27	27
	S	592	592	592	592	592	592	592	592	592	592	592	620	620	634	634	634	676	676
	ET	779	779	854	854	854	867	867	867	867	867	867	895	895	902	902	902	1008	1008
6000	UFU	60	60	60	60	60	60	60	60	60	60	60	80	80	80	80	80	80	80
	T	32	32	32	32	32	32	32	32	32	32	32	30	30	29	29	29	25	25
	S	592	592	592	592	592	592	592	592	592	592	592	620	620	630	630	630	642	642
	ET	779	779	854	854	854	854	867	867	867	867	867	881	895	900	900	900	991	991
5750	UFU	60	60	60	60	60	60	60	60	60	60	60	80	80	80	80	80	80	80
	T	30	30	30	30	30	30	30	30	30	30	30	29	29	28	28	28	26	26
	S	550	550	550	550	550	550	550	550	550	550	550	584	584	592	592	592	592	612
	ET	773	773	848	848	848	848	861	861	861	861	861	878	878	896	896	896	896	991
5500	UFU	60	60	60	60	60	60	60	60	60	60	60	60	60	80	80	80	80	80
	T	30	30	30	30	30	30	30	30	30	30	30	28	28	27	27	27	27	24
	S	530	530	530	530	530	530	530	530	530	530	530	580	580	580	592	592	592	612
	ET	763	763	763	838	838	838	838	851	851	851	851	876	876	876	896	896	896	991
5250	UFU	without	without	without	without	without	without	without	without	without	without	without	60	60	60	80	80	80	80
	T	29	29	29	29	29	29	29	29	29	29	29	27	27	27	26	26	26	23
	S	530	530	530	530	530	530	530	530	530	530	530	570	570	570	574	574	574	612
	ET	763	763	763	763	838	838	838	838	851	851	851	871	871	871	887	887	887	991
5000	UFU	without	without	without	without	without	without	without	without	without	without	without	60	60	60	60	60	60	80
	T	28	28	28	28	28	28	28	28	28	28	28	26	26	26	25	25	25	22
	S	530	530	530	530	530	530	530	530	530	530	530	570	570	570	568	568	568	612
	ET	763	763	763	763	838	838	838	838	851	851	851	871	871	871	883	883	883	991
4750	UFU	without	without	without	without	without	without	without	without	without	without	without	60	60	60	60	60	60	80
	T	26	26	26	26	26	26	26	26	26	26	26	25	25	25	24	24	24	21
	S	530	530	530	530	530	530	530	530	530	530	530	540	540	540	540	567	567	567
	ET	763	763	763	763	763	838	838	838	838	851	851	856	856	856	856	882	882	882
4500	UFU	without	without	without	without	without	without	without	without	without	without	without	without	without	40	40	60	60	60
	T	25	25	25	25	25	25	25	25	25	25	25	25	25	25	25	23	23	23
	S	510	510	510	510	510	510	510	510	510	510	510	540	540	540	540	566	566	566
	ET	753	753	753	753	753	828	828	828	828	828	841	856	856	856	856	881	881	881
4250	UFU	without	without	without	without	without	without	without	without	without	without	without	without	without	40	40	60	60	60
	T	25	25	25	25	25	25	25	25	25	25	25	23	23	23	23	22	22	22
	S	503	496	496	496	496	496	496	496	496	496	496	540	540	540	540	558	558	564
	ET	745	746	746	746	746	746	821	821	821	821	821	856	856	856	856	865	865	879
4000	UFU	without	without	without	without	without	without	without	without	without	without	without	without	without	40	40	60	60	60
	T	24	24	24	24	24	24	24	24	24	24	24	22	22	22	22	20	20	20
	S	503	503	496	496	496	496	496	496	496	496	496	540	540	540	540	540	538	538
	ET	745	745	746	746	746	746	746	821	821	821	821	856	856	856	856	856	855	855
3750	UFU	without	without	without	without	without	without	without	without	without	without	without	without	without	40	40	40	40	40
	T	22	22	22	22	22	22	22	22	22	22	22	20	20	20	20	20	20	20
	S	503	503	496	496	496	496	496	496	496	496	496	506	506	506	506	506	530	530
	ET	745	745	746	746	746	746	746	746	821	821	821	839	839	839	839	839	851	851
3500	UFU	without	without	without	without	without	without	without	without	without	without	without	without	without	40	40	40	40	40
	T	21	21	21	21	21	21	21	21	21	21	21	20	20	20	20	20	19	19
	S	490	490	490	470	470	470	470	470	470	470	470	506	506	506	506	506	524	524
	ET	732	732	732	733	733	733	733	733	733	808	808	826	839	839	839	839	848	848
3250	UFU	without	without	without	without	without	without	without	without	without	without	without	without	without	40	40	40	40	40
	T	20	20	20	20	20	20	20	20	20	20	20	19	19	19	19	19	18	18
	S	485	485	485	485	460	460	460	460	460	460	460	506	506	506	506	506	506	518
	ET	727	727	727	727	728	728	728	728	728	728	803	826	826	826	839	839	839	845
3000	UFU	without	without	without	without	without	without	without	without	without	without	without	without	without	40	40	40	40	40
	T	19	19	19	19	19	19	19	19	19	19	19	17	17	17	17	17	17	16
	S	485	485	485	485	485	460	460	460	460	460	460	496	496	496	496	496	496	504
	ET	727	727	727	727	727	728	728	728	728	728	728	821	821	821	821	834	834	838
2750	UFU	without	without	without	without	without	without	without	without	without	without	without	without	without	40	40	40	40	40
	T	17	17	17	17	17	17	17	17	17	17	17	16	16	16	16	16	16	15
	S	479	479	479	479	479	448	448	448	448	448	448	486	486	486	486	486	486	486
	ET	721	721	721	721	721	722	722	722	722	722	722	816	816	816	816	816	816	829
2500	UFU	without	without	without	without	without	without	without	without	without	without	without	without	without	40	40	40	40	40
	T	16	16	16	16	16	16	16	16	16	16	16	15	15	15	15	15	15	15
	S	470	470	470	470	470	470	470	430	430	430	430	486	486	486	486	486	486	486
	ET	712	712	712	712	712	712	712	713	713	713	713	741	816	816	816	816	816	816
2250	UFU	without	without	without	without	without	without	without	without	without	without	without	without	without	40	40	40	40	40
	T	15	15	15	15	15	15	15	15	15	15	15	13	13	13	13	13	13	13
	S	468	468	468	468	468	468	468	468	426	426	426	460	460	460	460	460	460	460
	ET	710	710	710	710	710	710	710	710	711	711	711	728	728	728	803	803	803	803
2000	UFU	without	without	without	without	without	without	without	without	without	without	without	without	without	40	40	40	40	40
	T	13	13	13	13	13	13	13	13	13	13	13	12	12	12	12	12	12	12
Width >		2000		2500		3000		3500		4000		4500		5000		5500		6000	

TKF 5.40
TAF 3.50
TAF 5.75
TAR 6.10
TAF 3.42
TAF 3.50
TAF 3.30
TAF 2.22
TAF 1.14

30. Table of reference values - rolling grating steel, TG-S and TG-E

Advice: The tables of standard values are only for standard shutters.

The following applies for the impulse control: The underfill height is always min. 40 mm.

219	273 mm								298 mm				323 mm			355 mm			
716	734	734	734	734	734	734	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	Height
626	644	644	644	644	644	644	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	
80	80	80	80	80	80	80	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	8000
32	29	29	29	29	29	29	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	
716	734	734	734	734	734	734	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	
626	644	644	644	644	644	644	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	7750
80	80	80	80	80	80	80	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	
31	29	29	29	29	29	29	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	
700	700	734	734	734	734	734	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	
610	610	644	644	644	644	644	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	7500
80	80	80	80	80	80	80	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	
31	31	28	28	28	28	28	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	
700	700	734	734	734	734	734	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	
610	610	644	644	644	644	644	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	7250
80	80	80	80	80	80	80	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	
30	30	27	27	27	27	27	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	
676	676	734	734	734	734	734	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	
586	586	644	644	644	644	644	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	7000
80	80	80	80	80	80	80	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	
30	30	26	26	26	26	26	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	
676	676	716	716	716	716	716	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	
586	586	626	626	626	626	626	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	6750
80	80	80	80	80	80	80	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	
29	29	25	25	25	25	25	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	
676	676	676	694	694	694	694	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	
586	586	586	604	604	604	604	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	6500
80	80	80	80	80	80	80	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	
28	28	28	25	25	25	25	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	
676	676	676	694	694	694	694	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	
586	586	586	604	604	604	604	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	6250
80	80	80	80	80	80	80	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	
27	27	27	24	24	24	24	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	
676	676	676	694	694	694	694	694	694	694	694	712	712	712	712	720	720	720	720	
1008	586	586	604	604	604	604	604	604	604	604	622	622	622	622	630	630	630	630	6000
80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	
25	25	25	23	23	23	23	23	23	23	23	22	22	22	22	21	21	21	21	
642	642	642	694	694	694	694	694	694	694	694	712	712	712	712	712	720	720	720	
991	991	575	604	604	604	604	604	604	604	604	622	622	622	622	622	630	630	630	5750
80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	
26	26	26	22	22	22	22	22	22	22	22	21	21	21	21	21	20	20	20	
612	612	612	612	664	664	664	664	664	664	664	664	664	682	682	682	670	670	670	
991	991	991	991	604	604	604	604	604	604	604	604	604	622	622	622	622	610	610	610	5500
80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	
24	24	24	24	21	21	21	21	21	21	21	21	21	20	20	20	20	20	20	
612	612	612	612	664	664	664	664	664	664	664	664	664	640	640	640	640	640	670	
991	991	991	991	604	604	604	604	604	604	604	604	604	580	580	580	580	610	610	5250
80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	
23	23	23	23	20	20	20	20	20	20	20	20	20	20	20	20	20	19	19	
612	612	612	612	612	622	622	622	622	622	622	622	622	640	640	640	640	670	670	
991	991	991	991	991	575	575	575	575	575	575	575	575	580	580	580	580	610	610	5000
80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	
22	22	22	22	22	20	20	20	20	20	20	20	20	19	19	19	19	18	18	
612	612	612	612	612	622	622	622	622	622	622	622	622	640	640	640	640	650	650	
991	991	991	991	991	575	575	575	575	575	575	575	575	580	580	580	580	590	590	4750
80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	
21	21	21	21	21	19	19	19	19	19	19	19	19	18	18	18	18	17	17	
612	612	612	612	612	622	622	622	622	622	622	622	622	640	640	640	640	640	650	
991	991	991	991	991	575	575	575	575	575	575	575	575	580	580	580	580	580	590	4500
80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	
20	20	20	20	20	18	18	18	18	18	18	18	18	17	17	17	17	17	16	
570	570	570	570	570	570	622	622	622	622	622	622	622	622	640	640	640	640	640	
885	885	970	970	970	970	575	575	575	575	575	575	575	575	580	580	580	580	580	4250
60	60	60	60	60	60	80	80	80	80	80	80	80	80	80	80	80	80	80	
20	20	20	20	20	20	17	17	17	17	17	17	17	17	16	16	16	16	16	
570	570	570	570	570	570	622	622	622	622	622	622	622	622	640	640	640	640	640	
885	885	885	970	970	970	575	575	575	575	575	575	575	575	580	580	580	580	580	4000
60	60	60	60	60	60	80	80	80	80	80	80	80	80	80	80	80	80	80	
19	19	19	19	19	19	16	16	16	16	16	16	16	16	15	15	15	15	15	
554	570	570	570	570	570	570	622	622	622	622	622	622	622	622	602	602	602	602	
869	885	885	885	885	970	970	575	575	575	575	575	575	575	575	575	575	575	575	3750
40	60	60	60	60	60	60	80	80	80	80	80	80	80	80	80	80	80	80	
20	18	18	18	18	18	18	15	15	15	15	15	15	15	15	15	15	15	15	
530	567	567	567	567	567	567	576	576	576	576	576	576	576	576	602	602	602	602	
851	882	882	882	882	882	882	575	575	575	575	575	575	575	575	575	575	575	575	3500
40	60	60	60	60	60	60	60	60	60	60	60	60	60	60	80	80	80	80	
19	16	16	16	16	16	16	15	15	15	15	15	15	15	15	14	14	14	14	
524	524	567	567	567	567	567	576	576	576	576	576	576	576	576	576	602	602	602	
848	848	882	882	882	882	882	973	973	973	575	575	575	575	575	575	575	575	575	3250
40	40	60	60	60	60	60	60	60	60	60	60	60	60	60	60	80	80	80	
18	18	15	15	15	15	15	14	14	14	14	14	14	14	14	14	13	13	13	
518	518	548	548	548	548	548	576	576	576	576	576	576	576	576	576	576	602	602	
845	845	863	863	863	863	863	973	973	973	575	575	575	575	575	575	575	575	575	3000
40	40	40	40	40	40	40	60	60	60	60	60	60	60	60	60	60	80	80	
16	16	15	15	15	15	15	13	13	13	13	13	13	13	13	13	13	12	12	
504	504	504	548	548	548	548	548	548	564	564	564	564	564	564	564	564	602	602	
838	838	838	863	863	863	863	863	863	967	967	967	967	967	575	575	575	575	575	2750
40	40	40	40	40	40	40	40	40	60	60	60	60	60	60	60	60	80	80	
15	15	15	14	14	14	14	14	14	12	12	12	12	12	12	12	12	11	11	
488	488	488	548	548	548	548	548	548	564	564	564	564	564	564	564	564	564	558	
830	830	830	863	863	863	863	863	863	967	967	967	967	967	967	967	967	575	575	2500
40	40	40	40	40	40	40	40	40	60	60	60	60	60	60	60	60	60	60	
14	14	14	12	12	12	12	12	12	11	11	11	11	11	11	11	11	11	10	
488	488	488	488	504	537	537	537	537	537	540	540	540	540	540	540	540	540	540	
830	830	830	830	838	852	852	852	852	852	955	955	955	955	955	955	955	955	955	2250
40	40	40	40	40	40	40	40	40	40	40	40	40	40	40	40	40	40	40	
13	13	13	13	12	12	12	12	12	12	10	10	10	10	10	10	10	10	10	
460	472	472	472	472	504	504	537	537	537	555	555	540	540	540	540	540	540	540	
803	822	822	822	822	838	838	852	852	852	870	870	955	955	955	955	955	955	955	2000
40	40	40	40	40	40	40	40	40	40	40	40	40	40	40	40	40	40	40	
12	12	12	12	12	10	10	10	10	10	9	9	9	9	9	9	9	9	9	
6500		7000		7500		8000		8500		9000		9500		10000		10500		11000	

TAR 6.100 TKR 7.97 TKR 6.60 TAR 5.100 TAR 5.75 TAR 3.42 TAF 4.50 TAF 3.50 TAR 6.100

31. Table of reference values - rolling grating Aluminium, TG-A and TG-V

S = Min. headroom
ET = Installation depth with emergency crank handle
UFU = Underfill height
T = Opening time of the shutter system approx. in seconds

Height		Shaft diameter 159 mm																	
	S	642	642	642	642	642	642	642	642	642	690	690	690	690	690	690	690	690	700
	ET	803	803	803	803	804	804	804	804	804	903	903	903	903	903	916	916	916	921
8000	UFU	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80
	T	39	39	39	39	39	39	39	39	39	36	36	36	36	36	36	36	36	35
	S	620	620	620	620	620	620	620	620	620	670	670	670	670	670	670	670	670	674
	ET	792	792	792	792	793	793	793	793	793	893	893	893	893	893	906	906	906	908
7750	UFU	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80
	T	39	39	39	39	39	39	39	39	39	36	36	36	36	36	36	36	36	35
	S	620	620	620	620	620	620	620	620	620	670	670	670	670	670	670	670	670	674
	ET	792	792	792	792	792	793	793	793	793	893	893	893	893	893	893	906	906	908
7500	UFU	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80
	T	38	38	38	38	38	38	38	38	38	35	35	35	35	35	35	35	35	34
	S	620	620	620	620	620	620	620	620	620	654	654	654	654	654	654	654	654	674
	ET	792	792	792	792	792	793	793	793	793	885	885	885	885	885	885	898	898	908
7250	UFU	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80
	T	37	37	37	37	37	37	37	37	37	35	35	35	35	35	35	35	35	33
	S	620	620	620	620	620	620	620	620	620	654	654	654	654	654	654	654	654	668
	ET	792	792	792	792	792	792	793	793	793	810	885	885	885	885	885	885	898	905
7000	UFU	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80
	T	35	35	35	35	35	35	35	35	35	33	33	33	33	33	33	33	33	32
	S	610	610	610	610	610	610	610	610	610	654	654	654	654	654	654	654	654	654
	ET	787	787	787	787	787	787	788	788	788	810	885	885	885	885	885	885	885	898
6750	UFU	60	60	60	60	60	60	60	60	60	80	80	80	80	80	80	80	80	80
	T	35	35	35	35	35	35	35	35	35	32	32	32	32	32	32	32	32	32
	S	592	592	592	592	592	592	592	592	592	648	648	648	648	648	648	648	648	648
	ET	778	778	778	778	778	778	779	779	779	807	807	882	882	882	882	882	882	882
6500	UFU	60	60	60	60	60	60	60	60	60	80	80	80	80	80	80	80	80	80
	T	34	34	34	34	34	34	34	34	34	31	31	31	31	31	31	31	31	31
	S	592	592	592	592	592	592	592	592	592	636	636	636	636	636	636	636	636	636
	ET	778	778	778	778	778	778	778	779	779	801	801	801	876	876	876	876	876	876
6250	UFU	60	60	60	60	60	60	60	60	60	80	80	80	80	80	80	80	80	80
	T	33	33	33	33	33	33	33	33	33	31	31	31	31	31	31	31	31	31
	S	592	592	592	592	592	592	592	592	592	620	620	620	620	620	620	620	620	620
	ET	778	778	778	778	778	778	778	778	779	793	793	793	793	868	868	868	868	868
6000	UFU	60	60	60	60	60	60	60	60	60	80	80	80	80	80	80	80	80	80
	T	32	32	32	32	32	32	32	32	32	30	30	30	30	30	30	30	30	30
	S	592	592	592	592	592	592	592	592	592	620	620	620	620	620	620	620	620	620
	ET	778	778	778	778	778	778	778	778	779	793	793	793	793	868	868	868	868	868
5750	UFU	60	60	60	60	60	60	60	60	60	80	80	80	80	80	80	80	80	80
	T	30	30	30	30	30	30	30	30	30	29	29	29	29	29	29	29	29	29
	S	550	550	550	550	550	550	550	550	550	584	584	584	584	584	584	584	584	584
	ET	772	772	772	772	772	772	772	772	772	790	790	790	790	790	865	865	865	865
5500	UFU	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60
	T	30	30	30	30	30	30	30	30	30	28	28	28	28	28	28	28	28	28
	S	530	530	530	530	530	530	530	530	530	580	580	580	580	580	580	580	580	580
	ET	762	762	762	762	762	762	762	762	762	788	788	788	788	788	788	863	863	863
5250	UFU	without	without	without	without	without	without	without	without	without	60	60	60	60	60	60	60	60	60
	T	29	29	29	29	29	29	29	29	29	27	27	27	27	27	27	27	27	27
	S	530	530	530	530	530	530	530	530	530	570	570	570	570	570	570	570	570	570
	ET	762	762	762	762	762	762	762	762	762	783	783	783	783	783	783	783	858	858
5000	UFU	without	without	without	without	without	without	without	without	without	60	60	60	60	60	60	60	60	60
	T	28	28	28	28	28	28	28	28	28	26	26	26	26	26	26	26	26	26
	S	530	530	530	530	530	530	530	530	530	570	570	570	570	570	570	570	570	570
	ET	762	762	762	762	762	762	762	762	762	783	783	783	783	783	783	783	783	858
4750	UFU	without	without	without	without	without	without	without	without	without	60	60	60	60	60	60	60	60	60
	T	26	26	26	26	26	26	26	26	26	25	25	25	25	25	25	25	25	25
	S	530	530	530	530	530	530	530	530	530	540	540	540	540	540	540	540	540	540
	ET	762	762	762	762	762	762	762	762	762	768	768	768	768	768	768	768	768	768
4500	UFU	without	without	without	without	without	without	without	without	without	without	without	without	without	40	40	40	40	40
	T	25	25	25	25	25	25	25	25	25	25	25	25	25	25	25	25	25	25
	S	510	510	510	510	510	510	510	510	510	540	540	540	540	540	540	540	540	540
	ET	752	752	752	752	752	752	752	752	752	767	768	768	768	768	768	768	768	768
4250	UFU	without	without	without	without	without	without	without	without	without	without	without	without	without	40	40	40	40	40
	T	25	25	25	25	25	25	25	25	25	23	23	23	23	23	23	23	23	23
	S	503	503	503	503	503	503	503	503	503	540	540	540	540	540	540	540	540	540
	ET	745	745	745	745	745	745	745	745	745	767	767	768	768	768	768	768	768	768
4000	UFU	without	without	without	without	without	without	without	without	without	without	without	without	without	40	40	40	40	40
	T	24	24	24	24	24	24	24	24	24	22	22	22	22	22	22	22	22	22
	S	503	503	503	503	503	503	503	503	503	540	540	540	540	540	540	540	540	540
	ET	745	745	745	745	745	745	745	745	745	767	767	767	768	768	768	768	768	768
3750	UFU	without	without	without	without	without	without	without	without	without	without	without	without	without	40	40	40	40	40
	T	22	22	22	22	22	22	22	22	22	20	20	20	20	20	20	20	20	20
	S	503	503	503	503	503	503	503	503	503	508	508	508	508	506	506	506	506	506
	ET	745	745	745	745	745	745	745	745	745	750	750	750	750	751	751	751	751	751
3500	UFU	without	without	without	without	without	without	without	without	without	without	without	without	without	40	40	40	40	40
	T	21	21	21	21	21	21	21	21	21	20	20	20	20	20	20	20	20	20
	S	490	490	490	490	490	490	490	490	490	508	508	508	508	508	506	506	506	506
	ET	732	732	732	732	732	732	732	732	732	750	750	750	750	750	751	751	751	751
3250	UFU	without	without	without	without	without	without	without	without	without	without	without	without	without	40	40	40	40	40
	T	20	20	20	20	20	20	20	20	20	19	19	19	19	19	19	19	19	19
	S	485	485	485	485	485	485	485	485	485	508	508	508	508	508	508	508	506	506
	ET	727	727	727	727	727	727	727	727	727	750	750	750	750	750	750	750	751	751
3000	UFU	without	without	without	without	without	without	without	without	without	without	without	without	without	40	40	40	40	40
	T	19	19	19	19	19	19	19	19	19	17	17	17	17	17	17	17	17	17
	S	485	485	485	485	485	485	485	485	485	503	503	503	503	503	503	503	503	503
	ET	727	727	727	727	727	727	727	727	727	745	745	745	745	745	745	745	745	745
2750	UFU	without	without	without	without	without	without	without	without	without	without	without	without	without	40	40	40	40	40
	T	17	17	17	17	17	17	17	17	17	16	16	16	16	16	16	16	16	16
	S	479	479	479	479	479	479	479	479	479	498	498	498	498	498	498	498	498	498
	ET	721	721	721	721	721	721	721	721	721	740	740	740	740	740	740	740	740	740
2500	UFU	without	without	without	without	without	without	without	without	without	without	without	without	without	40	40	40	40	40
	T	16	16	16	16	16	16	16	16	16	15	15	15	15	15	15	15	15	15
	S	470	470	470	470	470	470	470	470	470	498	498	498	498	498	498	498	498	498
	ET	712	712	712	712	712	712	712	712	712	740	740	740	740	740	740	740	740	740
2250	UFU	without	without	without	without	without	without	without	without	without	without	without	without	without	40	40	40	40	40
	T	15	15	15	15	15	15	15	15	15	13	13	13	13	13	13	13	13	13
	S	468	468	468	468	468	468	468	468	468	485	485	485	485	485	485	485	485	485
	ET	710	710	710	710	710	710	710	710	710	727	727	727	727	727	727	727	727	727
2000	UFU	without	without	without	without	without	without	without	without	without	without	without	without	without	40	40	40	40	40
	T	13	13	13	13	13	13	13	13	13	12	12	12	12	12	12	12	12	12
Width >		2000		2500		3000		3500		4000		4500		5000		5500		6000	

TAP 3.30 TAP 1.14 2.22

31. Table of reference values - rolling grating Aluminium, TG-A and TG-V

Advice: The tables of standard values are only for standard shutters.

The following applies for the impulse control: The underfill height is always min. 40 mm.

177 mm		219 mm						273 mm										298 mm		Height
700	700	716	716	716	716	716	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	
921	921	943	943	943	943	943	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	
80	80	80	80	80	80	80	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	8000
35	35	32	32	32	32	32	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	
674	674	716	716	716	716	716	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	
908	908	943	943	943	943	943	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	7750
80	80	80	80	80	80	80	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	
35	35	31	31	31	31	31	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	
674	674	674	700	700	700	700	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	
908	908	908	935	935	935	935	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	7500
80	80	80	80	80	80	80	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	
34	34	34	31	31	31	31	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	
674	674	674	700	700	700	700	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	
908	908	908	935	935	935	935	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	7250
80	80	80	80	80	80	80	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	
33	33	33	30	30	30	30	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	
668	668	668	676	676	676	676	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	
905	905	905	923	923	923	923	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	7000
80	80	80	80	80	80	80	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	
32	32	32	30	30	30	30	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	
668	668	668	676	676	676	676	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	
905	905	905	923	923	923	923	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	6750
80	80	80	80	80	80	80	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	
31	31	31	29	29	29	29	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	
648	648	648	676	676	676	676	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	
895	895	895	923	923	923	923	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	6500
80	80	80	80	80	80	80	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	
31	31	31	28	28	28	28	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	
634	634	634	634	676	676	676	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	
888	888	888	888	923	923	923	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	6250
80	80	80	80	80	80	80	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	
30	30	30	30	27	27	27	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	-/-	
620	634	634	634	676	676	676	676	676	676	694	694	694	694	694	694	694	694	694	694	
868	888	888	888	909	909	923	923	923	923	932	1017	1017	1017	1017	1017	1017	1017	1017	1017	6000
80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	
30	29	29	29	25	25	25	25	25	25	23	23	23	23	23	23	23	23	23	23	
620	630	630	630	642	642	642	642	642	642	642	694	694	694	694	694	694	694	694	694	
868	886	886	886	892	892	892	906	906	906	906	932	932	1017	1017	1017	1017	1017	1017	1017	5750
80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	
29	28	28	28	26	26	26	26	26	26	26	22	22	22	22	22	22	22	22	22	
584	592	592	592	592	612	612	612	612	612	612	664	664	664	664	664	664	664	664	664	
865	882	882	882	882	892	892	892	906	906	906	932	932	932	1017	1017	1017	1017	1017	1017	5500
60	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	
28	27	27	27	27	24	24	24	24	24	24	21	21	21	21	21	21	21	21	21	
580	592	592	592	592	612	612	612	612	612	612	664	664	664	664	664	664	664	664	664	
863	882	882	882	882	892	892	892	892	906	906	932	932	932	932	932	1017	1017	1017	1017	5250
60	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	
27	26	26	26	26	23	23	23	23	23	23	20	20	20	20	20	20	20	20	20	
570	574	574	574	574	612	612	612	612	612	612	622	622	622	622	622	622	622	622	622	
858	860	873	873	873	892	892	892	892	892	892	911	911	911	911	911	911	911	911	996	5000
60	60	60	60	60	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	
26	25	25	25	25	22	22	22	22	22	22	20	20	20	20	20	20	20	20	20	
570	566	566	566	566	612	612	612	612	612	612	612	622	622	622	622	622	622	622	622	
858	856	856	869	869	892	892	892	892	892	892	892	911	911	911	911	911	911	911	911	4750
60	60	60	60	60	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	
25	24	24	24	24	21	21	21	21	21	21	21	19	19	19	19	19	19	19	19	
540	540	564	564	564	612	612	612	612	612	612	612	622	622	622	622	622	622	622	622	
768	843	855	855	868	892	892	892	892	892	892	892	911	911	911	911	911	911	911	911	4500
40	40	60	60	60	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	
25	25	23	23	23	20	20	20	20	20	20	20	18	18	18	18	18	18	18	18	
540	540	562	562	562	562	570	570	570	570	570	570	570	622	622	622	622	622	622	622	
768	843	854	854	854	867	871	871	871	871	871	871	871	911	911	911	911	911	911	911	4250
40	40	60	60	60	60	60	60	60	60	60	60	60	80	80	80	80	80	80	80	
23	23	22	22	22	22	20	20	20	20	20	20	20	17	17	17	17	17	17	17	
540	540	540	558	558	558	570	570	570	570	570	570	570	622	622	622	622	622	622	622	
768	768	843	852	852	852	871	871	871	871	871	871	871	911	911	911	911	911	911	911	4000
40	40	40	60	60	60	60	60	60	60	60	60	60	80	80	80	80	80	80	80	
22	22	22	20	20	20	19	19	19	19	19	19	19	16	16	16	16	16	16	16	
540	540	540	538	538	538	538	570	570	570	570	570	570	622	622	622	622	622	622	622	
768	768	768	842	842	842	842	871	871	871	871	871	871	911	911	911	911	911	911	911	3750
40	40	40	40	40	40	40	60	60	60	60	60	60	80	80	80	80	80	80	80	
20	20	20	20	20	20	20	18	18	18	18	18	18	15	15	15	15	15	15	15	
506	506	506	530	530	530	530	564	564	564	564	564	564	564	576	576	576	576	576	576	
751	751	751	838	838	838	838	868	868	868	868	868	868	868	888	888	888	888	888	888	3500
40	40	40	40	40	40	40	60	60	60	60	60	60	60	60	60	60	60	60	60	
20	20	20	19	19	19	19	16	16	16	16	16	16	16	15	15	15	15	15	15	
506	506	506	524	524	524	524	564	564	564	564	564	564	564	576	576	576	576	576	576	
751	751	751	835	835	835	835	868	868	868	868	868	868	868	874	888	888	888	888	888	3250
40	40	40	40	40	40	40	60	60	60	60	60	60	60	60	60	60	60	60	60	
19	19	19	18	18	18	18	15	15	15	15	15	15	15	14	14	14	14	14	14	
506	506	506	506	518	518	518	518	526	526	526	526	526	526	526	576	576	576	576	576	
751	751	751	751	832	832	832	832	849	849	849	849	849	849	849	874	874	888	888	888	3000
40	40	40	40	40	40	40	40	40	40	40	40	40	40	40	60	60	60	60	60	
17	17	17	17	16	16	16	16	15	15	15	15	15	15	15	13	13	13	13	13	
496	496	496	496	504	504	504	504	526	526	526	526	526	526	526	564	564	564	564	564	
746	746	746	746	750	750	750	825	836	849	849	849	849	849	849	868	868	868	868	868	2750
40	40	40	40	40	40	40	40	40	40	40	40	40	40	40	60	60	60	60	60	
16	16	16	16	15	15	15	15	14	14	14	14	14	14	14	12	12	12	12	12	
498	486	486	486	486	488	488	488	488	526	526	526	526	526	526	564	564	564	564	564	
740	741	741	741	741	742	742	742	742	836	836	849	849	849	849	868	868	868	868	868	2500
40	40	40	40	40	40	40	40	40	40	40	40	40	40	40	60	60	60	60	60	
15	15	15	15	15	14	14	14	14	12	12	12	12	12	12	11	11	11	11	11	
498	486	486	486	486	488	488	488	488	504	504	504	504	504	504	504	540	540	540	540	
740	741	741	741	741	742	742	742	742	825	825	825	825	825	838	838	856	856	856	856	2250
40	40	40	40	40	40	40	40	40	40	40	40	40	40	40	40	40	40	40	40	
13	13	13	13	13	13	13	13	13	12	12	12	12	12	12	12	10	10	10	10	
485	485	485	485	460	460	472	472	472	504	504	504	504	504	504	504	504	540	540	540	
727	727	727	727	728	728	734	734	734	825	825	825	825	825	825	825	825	856	856	856	2000
40	40	40	40	40	40	40	40	40	40	40	40	40	40	40	40	40	40	40	40	
12	12	12	12	12	12	12	12	12	10	10	10	10	10	10	10	10	9	9	9	
6500		7000		7500		8000		8500		9000		9500		10000		10500		11000		

TAR 5.75
TAR 6.10
TAR 3.42
TAR 3.50
TAR 3.56
TAR 3.36

32. Table of reference values - rolling grating Aluminium, TG-X

S = Min. headroom
ET = Installation depth with emergency crank handle
UFU = Underfill height
T = Opening time of the shutter system approx. in seconds

Height		Shaft diameter 159 mm																	
	S	600	600	600	600	600	600	600	600	600	600	600	600	600	600	600	600	600	600
	ET	782	782	782	782	782	782	782	783	783	783	783	783	783	783	783	858	858	858
6000	UFU	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60
	T	31	31	31	31	31	31	31	31	31	31	31	31	31	31	31	31	31	31
	S	600	600	600	600	600	600	600	600	600	600	600	600	600	600	600	600	600	600
	ET	782	782	782	782	782	782	782	783	783	783	783	783	783	783	783	783	858	858
5750	UFU	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60
	T	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30
	S	570	570	570	570	570	570	570	570	570	570	570	570	570	570	570	570	570	570
	ET	782	782	782	782	782	782	782	782	783	783	783	783	783	783	783	783	783	858
5500	UFU	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60
	T	29	29	29	29	29	29	29	29	29	29	29	29	29	29	29	29	29	29
	S	560	560	560	560	560	560	560	560	560	560	560	560	560	560	560	560	560	560
	ET	777	777	777	777	777	777	777	777	777	778	778	778	778	778	778	778	778	778
5250	UFU	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60
	T	28	28	28	28	28	28	28	28	28	28	28	28	28	28	28	28	28	28
	S	560	560	560	560	560	560	560	560	560	560	560	560	560	560	560	560	560	560
	ET	777	777	777	777	777	777	777	777	777	778	778	778	778	778	778	778	778	778
5000	UFU	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60
	T	26	26	26	26	26	26	26	26	26	26	26	26	26	26	26	26	26	26
	S	560	560	560	560	560	560	560	560	560	560	560	560	560	560	560	560	560	560
	ET	777	777	777	777	777	777	777	777	777	777	778	778	778	778	778	778	778	778
4750	UFU	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60
	T	25	25	25	25	25	25	25	25	25	25	25	25	25	25	25	25	25	25
	S	538	538	538	538	538	538	538	538	538	538	538	538	538	538	538	538	538	538
	ET	766	766	766	766	766	766	766	766	766	766	766	767	767	767	767	767	767	767
4500	UFU	without	without	without	without	without	without	without	without	without	without	without	without	without	40	40	40	40	40
	T	25	25	25	25	25	25	25	25	25	25	25	25	25	25	25	25	25	25
	S	538	538	538	538	538	538	538	538	538	538	538	538	538	538	538	538	538	538
	ET	766	766	766	766	766	766	766	766	766	766	766	766	767	767	767	767	767	767
4250	UFU	without	without	without	without	without	without	without	without	without	without	without	without	without	40	40	40	40	40
	T	23	23	23	23	23	23	23	23	23	23	23	23	23	23	23	23	23	23
	S	538	538	538	538	538	538	538	538	538	538	538	538	538	538	538	538	538	538
	ET	766	766	766	766	766	766	766	766	766	766	766	766	766	767	767	767	767	767
4000	UFU	without	without	without	without	without	without	without	without	without	without	without	without	without	40	40	40	40	40
	T	22	22	22	22	22	22	22	22	22	22	22	22	22	22	22	22	22	22
	S	505	505	505	505	505	505	505	505	505	505	505	505	505	505	505	500	500	500
	ET	747	747	747	747	747	747	747	747	747	747	747	747	747	747	747	748	748	748
3750	UFU	without	without	without	without	without	without	without	without	without	without	without	without	without	40	40	40	40	40
	T	22	22	22	22	22	22	22	22	22	22	22	22	22	22	22	22	22	22
	S	505	505	505	505	505	505	505	505	505	505	505	505	505	505	505	505	500	500
	ET	747	747	747	747	747	747	747	747	747	747	747	747	747	747	747	747	748	748
3500	UFU	without	without	without	without	without	without	without	without	without	without	without	without	without	40	40	40	40	40
	T	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20
	S	505	505	505	505	505	505	505	505	505	505	505	505	505	505	505	505	505	500
	ET	747	747	747	747	747	747	747	747	747	747	747	747	747	747	747	747	747	748
3250	UFU	without	without	without	without	without	without	without	without	without	without	without	without	without	40	40	40	40	40
	T	19	19	19	19	19	19	19	19	19	19	19	19	19	19	19	19	19	19
	S	483	483	483	483	483	483	483	483	483	483	483	483	483	483	483	483	483	483
	ET	725	725	725	725	725	725	725	725	725	725	725	725	725	725	725	725	725	725
3000	UFU	without	without	without	without	without	without	without	without	without	without	without	without	without	40	40	40	40	40
	T	19	19	19	19	19	19	19	19	19	19	19	19	19	19	19	19	19	19
	S	483	483	483	483	483	483	483	483	483	483	483	483	483	483	483	483	483	483
	ET	725	725	725	725	725	725	725	725	725	725	725	725	725	725	725	725	725	725
2750	UFU	without	without	without	without	without	without	without	without	without	without	without	without	without	40	40	40	40	40
	T	17	17	17	17	17	17	17	17	17	17	17	17	17	17	17	17	17	17
	S	483	483	483	483	483	483	483	483	483	483	483	483	483	483	483	483	483	483
	ET	725	725	725	725	725	725	725	725	725	725	725	725	725	725	725	725	725	725
2500	UFU	without	without	without	without	without	without	without	without	without	without	without	without	without	40	40	40	40	40
	T	16	16	16	16	16	16	16	16	16	16	16	16	16	16	16	16	16	16
	S	470	470	470	470	470	470	470	470	470	470	470	470	470	470	470	470	470	470
	ET	712	712	712	712	712	712	712	712	712	712	712	712	712	712	712	712	712	712
2250	UFU	without	without	without	without	without	without	without	without	without	without	without	without	without	40	40	40	40	40
	T	15	15	15	15	15	15	15	15	15	15	15	15	15	15	15	15	15	15
	S	470	470	470	470	470	470	470	470	470	470	470	470	470	470	470	470	470	470
	ET	712	712	712	712	712	712	712	712	712	712	712	712	712	712	712	712	712	712
2000	UFU	without	without	without	without	without	without	without	without	without	without	without	without	without	40	40	40	40	40
	T	13	13	13	13	13	13	13	13	13	13	13	13	13	13	13	13	13	13
Width >		2000		2500		3000		3500		4000		4500		5000		5500		6000	

TAP 2.22
TAP 1.14

32. Table of reference values - rolling grating Aluminium, TG-X

Advice: The tables of standard values are only for standard shutters.

The following applies for the impulse control: The underfill height is always min. 40 mm.

159 mm	177 mm				219 mm							273 mm							
600	624	624	624	624	630	630	630	630	630	630	630	650	650	650	650	650	650	650	Height
858	870	883	883	883	886	886	886	886	886	886	900	910	910	910	910	910	910	995	
60	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	9000
31	29	29	29	29	27	27	27	27	27	27	27	25	25	25	25	25	25	25	
600	590	590	590	590	630	630	630	630	630	630	630	650	650	650	650	650	650	650	
858	853	853	866	866	886	886	886	886	886	886	886	910	910	910	910	910	910	910	5750
60	60	60	60	60	80	80	80	80	80	80	80	80	80	80	80	80	80	80	
30	30	30	30	30	26	26	26	26	26	26	26	23	23	23	23	23	23	23	
570	570	560	560	560	560	580	580	580	580	580	580	620	620	620	620	620	620	620	
858	858	853	853	866	866	876	876	876	876	876	876	910	910	910	910	910	910	910	5500
60	60	60	60	60	60	60	60	60	60	60	60	80	80	80	80	80	80	80	
29	29	28	28	28	28	26	26	26	26	26	26	22	22	22	22	22	22	22	
560	560	560	560	560	560	580	580	580	580	580	580	580	594	594	594	594	594	594	
853	853	853	853	853	853	876	876	876	876	876	876	876	897	897	897	897	897	897	5250
60	60	60	60	60	60	60	60	60	60	60	60	60	80	80	80	80	80	80	
28	28	27	27	27	27	24	24	24	24	24	24	24	22	22	22	22	22	22	
560	560	560	560	560	560	580	580	580	580	580	580	580	594	594	594	594	594	594	
778	853	853	853	853	853	876	876	876	876	876	876	876	897	897	897	897	897	897	5000
60	60	60	60	60	60	60	60	60	60	60	60	60	80	80	80	80	80	80	
26	26	26	26	26	26	23	23	23	23	23	23	23	21	21	21	21	21	21	
560	560	560	550	550	550	580	580	580	580	580	580	580	594	594	594	594	594	594	
778	778	853	848	848	848	848	876	876	876	876	876	876	897	897	897	897	897	897	4750
60	60	60	60	60	60	60	60	60	60	60	60	60	80	80	80	80	80	80	
25	25	25	25	25	25	22	22	22	22	22	22	22	20	20	20	20	20	20	
538	538	538	550	550	550	550	556	556	556	556	556	556	594	594	594	594	594	594	
767	767	767	848	848	848	848	864	864	864	864	864	864	897	897	897	897	897	897	4500
40	40	40	60	60	60	60	60	60	60	60	60	60	80	80	80	80	80	80	
25	25	25	23	23	23	23	22	22	22	22	22	22	19	19	19	19	19	19	
538	538	538	550	550	550	550	556	556	556	556	556	556	556	578	578	578	578	578	
767	767	767	848	848	848	848	864	864	864	864	864	864	864	889	889	889	889	889	4250
40	40	40	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	
23	23	23	22	22	22	22	20	20	20	20	20	20	20	18	18	18	18	18	
538	538	538	538	506	506	506	556	556	556	556	556	556	556	578	578	578	578	578	
767	767	767	767	826	826	826	864	864	864	864	864	864	864	875	875	889	889	889	4000
40	40	40	40	40	40	40	60	60	60	60	60	60	60	60	60	60	60	60	
22	22	22	22	22	22	22	19	19	19	19	19	19	19	17	17	17	17	17	
500	500	500	500	506	506	506	506	556	556	556	556	556	556	556	578	578	578	578	
748	748	748	748	751	826	826	826	864	864	864	864	864	864	864	875	875	875	889	3750
40	40	40	40	40	40	40	40	60	60	60	60	60	60	60	60	60	60	60	
22	22	22	22	21	21	21	21	18	18	18	18	18	18	18	16	16	16	16	
500	500	500	500	506	506	506	506	536	536	536	536	536	536	536	578	578	578	578	
748	748	748	748	751	751	751	826	841	841	854	854	854	854	854	875	875	875	875	3500
40	40	40	40	40	40	40	40	40	40	40	40	40	40	40	60	60	60	60	
20	20	20	20	20	20	20	20	17	17	17	17	17	17	17	15	15	15	15	
500	500	500	500	500	506	506	506	506	536	536	536	536	536	536	560	560	560	560	
748	748	748	748	748	751	751	751	751	841	841	841	854	854	854	866	866	866	866	3250
40	40	40	40	40	40	40	40	40	40	40	40	40	40	40	60	60	60	60	
19	19	19	19	19	18	18	18	18	16	16	16	16	16	16	14	14	14	14	
483	456	456	456	456	496	496	496	496	536	536	536	536	536	536	536	536	560	560	
725	726	726	726	726	746	746	746	746	841	841	841	841	841	854	854	866	866	866	3000
40	40	40	40	40	40	40	40	40	40	40	40	40	40	40	40	60	60	60	
19	19	19	19	19	17	17	17	17	15	15	15	15	15	15	15	13	13	13	
483	483	483	456	456	456	496	496	496	526	526	526	526	526	526	526	560	560	560	
725	725	725	726	726	726	746	746	746	836	836	836	836	836	836	836	866	866	866	2750
40	40	40	40	40	40	40	40	40	40	40	40	40	40	40	40	60	60	60	
17	17	17	17	17	17	16	16	16	14	14	14	14	14	14	14	12	12	12	
483	483	483	483	483	456	496	496	496	496	526	526	526	526	526	526	526	530	530	
725	725	725	725	725	726	746	746	746	746	836	836	836	836	836	836	836	851	851	2500
40	40	40	40	40	40	40	40	40	40	40	40	40	40	40	40	40	40	40	
16	16	16	16	16	16	14	14	14	14	12	12	12	12	12	12	12	11	11	
470	470	470	470	470	470	470	460	460	460	526	526	526	526	526	526	526	530	530	
712	712	712	712	712	712	712	728	728	728	761	836	836	836	836	836	836	851	851	2250
40	40	40	40	40	40	40	40	40	40	40	40	40	40	40	40	40	40	40	
15	15	15	15	15	15	15	14	14	14	11	11	11	11	11	11	11	10	10	
470	470	470	470	470	470	470	460	460	460	460	486	486	486	486	486	486	486	530	
712	712	712	712	712	712	712	728	728	728	728	741	741	741	741	816	816	816	851	2000
40	40	40	40	40	40	40	40	40	40	40	40	40	40	40	40	40	40	40	
13	13	13	13	13	13	13	12	12	12	12	11	11	11	11	11	11	11	9	
6500		**7000**		**7500**		**8000**		**8500**		**9000**		**9500**		**10000**		**10500**		**11000**	

TAF 3.50 TAF 3.30 TAF 3.42 TAF 5.75 TAF 3.50 TAF 2.22 TAF 6.30

33. Table of reference values - 6010 Aluminium visio

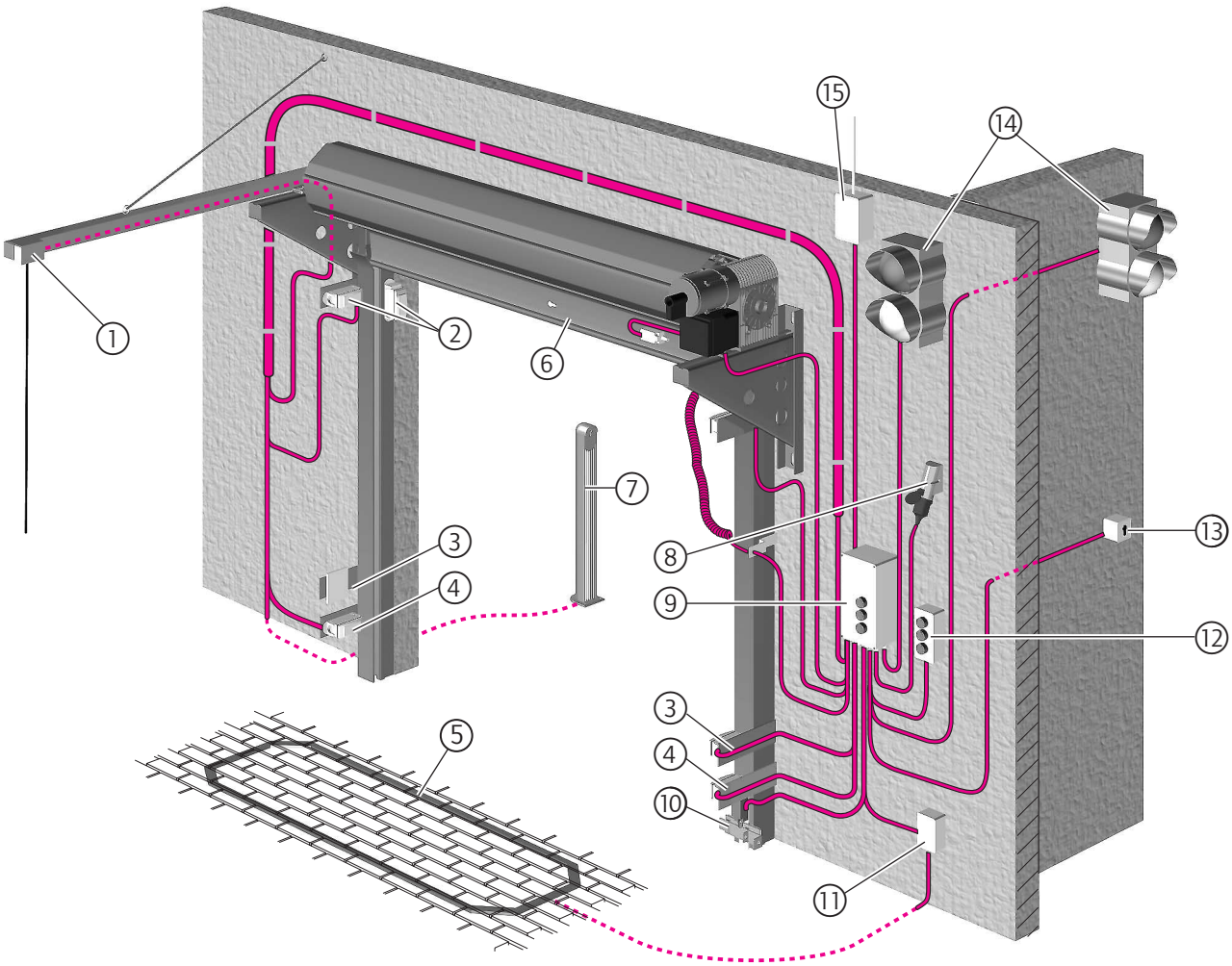
S = Min. headroom
ET = Installation depth with emergency crank handle
UFU = Underfill height
T = Opening time of the shutter system approx. in seconds

Height		Shaft diameter 159 mm															Height
6000	S	638	638	638	638	638	638	638	638	638	638	638	638	638	638	638	6000
	ET	801	801	801	801	801	801	801	801	801	801	801	801	801	801	802	
	UFU	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	
	T	29	29	29	29	29	29	29	29	29	29	29	29	29	29	29	
5750	S	638	638	638	638	638	638	638	638	638	638	638	638	638	638	638	5750
	ET	801	801	801	801	801	801	801	801	801	801	801	801	801	801	801	
	UFU	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	
	T	28	28	28	28	28	28	28	28	28	28	28	28	28	28	28	
5500	S	608	608	608	608	608	608	608	608	608	608	608	608	608	608	608	5500
	ET	801	801	801	801	801	801	801	801	801	801	801	801	801	801	801	
	UFU	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	
	T	27	27	27	27	27	27	27	27	27	27	27	27	27	27	27	
5250	S	570	570	570	570	570	570	570	570	570	570	570	570	570	570	570	5250
	ET	782	782	782	782	782	782	782	782	782	782	782	782	782	782	782	
	UFU	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	
	T	27	27	27	27	27	27	27	27	27	27	27	27	27	27	27	
5000	S	570	570	570	570	570	570	570	570	570	570	570	570	570	570	570	5000
	ET	782	782	782	782	782	782	782	782	782	782	782	782	782	782	782	
	UFU	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	
	T	26	26	26	26	26	26	26	26	26	26	26	26	26	26	26	
4750	S	570	570	570	570	570	570	570	570	570	570	570	570	570	570	570	4750
	ET	782	782	782	782	782	782	782	782	782	782	782	782	782	782	782	
	UFU	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	
	T	25	25	25	25	25	25	25	25	25	25	25	25	25	25	25	
4500	S	570	570	570	570	570	570	570	570	570	570	570	570	570	570	570	4500
	ET	782	782	782	782	782	782	782	782	782	782	782	782	782	782	782	
	UFU	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	
	T	23	23	23	23	23	23	23	23	23	23	23	23	23	23	23	
4250	S	524	524	524	524	524	524	524	524	524	524	524	524	524	524	524	4250
	ET	759	759	759	759	759	759	759	759	759	759	759	759	759	759	759	
	UFU	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	
	T	24	24	24	24	24	24	24	24	24	24	24	24	24	24	24	
4000	S	524	524	524	524	524	524	524	524	524	524	524	524	524	524	524	4000
	ET	759	759	759	759	759	759	759	759	759	759	759	759	759	759	759	
	UFU	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	
	T	22	22	22	22	22	22	22	22	22	22	22	22	22	22	22	
3750	S	524	524	524	524	524	524	524	524	524	524	524	524	524	524	524	3750
	ET	759	759	759	759	759	759	759	759	759	759	759	759	759	759	759	
	UFU	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	
	T	21	21	21	21	21	21	21	21	21	21	21	21	21	21	21	
3500	S	524	524	524	524	524	524	524	524	524	524	524	524	524	524	524	3500
	ET	759	759	759	759	759	759	759	759	759	759	759	759	759	759	759	
	UFU	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	
	T	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	
3250	S	498	498	498	498	498	498	498	498	498	498	498	498	498	498	498	3250
	ET	740	740	740	740	740	740	740	740	740	740	740	740	740	740	740	
	UFU	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	
	T	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	
3000	S	498	498	498	498	498	498	498	498	498	498	498	498	498	498	498	3250
	ET	740	740	740	740	740	740	740	740	740	740	740	740	740	740	740	
	UFU	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	
	T	18	18	18	18	18	18	18	18	18	18	18	18	18	18	18	
2750	S	498	498	498	498	498	498	498	498	498	498	498	498	498	498	498	2750
	ET	740	740	740	740	740	740	740	740	740	740	740	740	740	740	740	
	UFU	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	
	T	16	16	16	16	16	16	16	16	16	16	16	16	16	16	16	
2500	S	498	498	498	498	498	498	498	498	498	498	498	498	498	498	498	2500
	ET	740	740	740	740	740	740	740	740	740	740	740	740	740	740	740	
	UFU	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	
	T	15	15	15	15	15	15	15	15	15	15	15	15	15	15	15	
2250	S	478	478	478	478	478	478	478	478	478	478	478	478	478	478	478	2250
	ET	720	720	720	720	720	720	720	720	720	720	720	720	720	720	720	
	UFU	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	
	T	15	15	15	15	15	15	15	15	15	15	15	15	15	15	15	
2000	S	478	478	478	478	478	478	478	478	478	478	478	478	478	478	478	2000
	ET	720	720	720	720	720	720	720	720	720	720	720	720	720	720	720	
	UFU	without	without	without	without	without	without	without	without	without	without	without	without	without	without	without	
	T	13	13	13	13	13	13	13	13	13	13	13	13	13	13	13	
Width >		2000		2500		3000		3500		4000		4500		5000		5500	

Shutters for indoors, resistance to wind load class 2 in accordance with DIN EN 13241-1

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34. Wiring plan



Legend wiring plan

No.	Name		Voltage (V)	Wire cross section	Max. length (m)
1	Cantilever with pull switch		24	2 x 0.5 mm²	25
2	Roll-up safety device	Transmitter (3 m incl.)	24	2 x 0.5 mm²	25
		Receiver (12 m incl.)	24	4 x 0.5 mm²	25
3	Clearance photo switch (reflex)		24	4 x 0.5 mm²	25
4	Clearance photo switch (one-way)	Transmitter (3 m incl.)	24	2 x 0.5 mm²	25
		Receiver (12 m incl.)	24	4 x 0.5 mm²	25
5	Induction loop		–	–	–
6	Lockable bottom profile		24	2 x 0.5 mm²	25
7	Aluminium column with key-operated switch		24	2 x 0.5 mm² (per contact)	25
8	Socket 16 A / 5-wire		–	–	–
9	Control		–	Pre-wired as standard	–
10	Sliding bolt		24	2 x 0.5 mm²	25
11	Evaluation unit for induction loop (feed cable)		230	3 x 1.5 mm²	25
12	Push button		24	2 x 0.5 mm² (per contact)	25
13	Key-operated switch	OPEN – CLOSE	24	3 x 0.5 mm²	25
		OPEN – CLOSE with stop button	24	5 x 0.5 mm²	25
14	Traffic light		230	3 x 1.5 mm²	50
15	External radio receiver		24	6 x 0.5 mm²	25

Subject to technical changes
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